Asymmetry effects in homodyne-like measurements: Positive operator-valued measures and quantum key distribution

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We study applicability of the Gaussian approximation describing photon count statistics for both the homodyne and the double homodyne measurements in the presence of asymmetry effects when the beam splitters are unbalanced and the quantum efficiencies of the photodetectors are not identical. We also use the Gaussian approximation to construct the positive operator-valued measure (POVM) that takes into account the asymmetry effects. The results are applied to calculate the asymptotic secret fraction of the GG02 protocol under an untrusted noise model. It is found that asymmetry is detrimental in this case. Moreover, we found that the double homodyne POVM is not uniquely defined, because the squeezing parameter, which governs the POVM, is constrained within an interval determined by the asymmetry parameters of the measurement scheme. Consequently, the Holevo information, which depends on this POVM, also varies with the squeezing parameter within this range.

I. INTRODUCTION

Homodyne and double homodyne detection are important measurement methods in quantum optics, widely used in continuous-variable quantum key distribution (CV-QKD) protocols. These techniques measure quadratures of light, enabling the receiver to retrieve the information encoded in the transmitted signals, which is essential for secure secret key generation in CV-QKD protocols [1–4]. Homodyne detection measures one quadrature by mixing a weak quantum signal with a strong local oscillator (LO) at a beam splitter, while double homodyne detection measures both conjugate quadratures at once, giving a fuller picture of the quantum state [5, 6].

In a typical homodyne setup, the quantum signal and classical LO combine at a beam splitter, and two photodiodes record the outputs. The difference in their photocurrents reveals the quadrature information [7, 8]. Most theoretical treatments assume the LO is much stronger than the signal, which simplifies calculations [9]. However, this can hide some quantum effects and increase classical noise. When LO and signal have similar strength, phase-sensitive quantum features become clearer, but this requires highly efficient, stable detectors [10, 11]. Other measurement setups use intensity correlations or unbalanced beam splitters to help when detection efficiency is low [10].

Double homodyne (eight-port) detection measures both quadratures simultaneously, allowing direct reconstruction of the Husimi Q-function. This setup works well at various LO intensities, bridging classical and quantum regimes [5, 12, 13]. In CV-QKD, it helps symmetrize measurements by keeping full quadrature data [6].

Both homodyne and double homodyne detection are examples of Gaussian measurements. Their Gaussian nature is beneficial for analyzing and processing quantum signals in CV-QKD using well-established mathematical tools. In contrast, other approaches such as unbalanced homodyne detection, which measures only one output port of the beam splitter, realize non-Gaussian measurements based on positive operator-valued measures (POVMs) associated with s-parametrized quasiprobability distributions ($s < 1$) [14]. These schemes suffer from higher noise sensitivity and current limitations in photon-number resolution, although detector arrays have been proposed to mitigate these drawbacks [12, 15, 16], making their analysis more complex compared to Gaussian homodyne measurements.

Practical implementations of homodyne-based measurements are inevitably affected by nonideal components and asymmetries. Imperfections such as unbalanced beam splitters, unequal detector efficiencies, limited photon-number resolution, and detector dead times introduce structured excess noise that degrades the fidelity of measurements and compromises the security of CV-QKD systems [14, 17–22]. While the ideal POVMs associated with projections onto quadrature or coherent states have been thoroughly studied [5, 9], a comprehensive quantum-optical framework explicitly incorporating beam splitter imbalance and detector-efficiency mismatch into the measurement operators remains absent [15, 16]. Existing complementary studies, such as Ref. [23], have investigated the practical influence of detection imbalance on phase-reference estimation and key-rate performance in CV-QKD systems and illustrated

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its impact experimentally using Wigner-function reconstructions. However, these analyses remain semi-classical and do not address the modification of the measurement process itself within the quantum-optical formalism. In practice, these imperfections are often modeled as additive technical noise (electronic noise, dark counts, finite bandwidth), but asymmetry introduces structured excess noise that must be accounted for to produce reliable security estimates [21–24]. Such imperfections may also open side channels exploitable by an adversary (e.g., wavelength-dependent or detector-blinding attacks), so countermeasures such as spectral filtering, detector balancing, and careful calibration are essential [22, 25–28].

In this work, we focus on these imperfections at the level of the quantum measurement and refer to them collectively as asymmetry effects. Specifically, our analysis focuses on an asymmetrical detection scenario: an unbalanced beam splitter combined with unequal quantum efficiencies of the photodetectors, in contrast to the conventional symmetrical case of a balanced beam splitter and identical detector efficiencies. To derive tractable expressions, we model the photocount statistics by approximating Poisson distributions with Gaussian ones (the Gaussian approximation) and further simplify using the strong LO approximation. From these approximations, we construct the corresponding POVMs.We then apply the developed formalism to a homodyne-based two-quadrature scheme (double homodyne, often referred to as heterodyne in the QKD literature [3, 4, 29]) to demonstrate applicability beyond single-quadrature detection. We find that the asymmetrical double homodyne POVM requires an extension to the set of squeezed coherent states. Interpreting those POVMs as classical mixing channels [30] in control of adversary (untrusted noise scenario [24]), we utilize them to compute Holevo information in the Gaussian-modulated coherent-state (GMCS or GG02) protocol [31]. Then, after calculating mutual information, we are able to determine asymptotic secret fraction, thereby quantifying the impact of measurement asymmetry on protocol security.

The paper is organized as follows. In Sec. II, we derive the statistical distribution of difference photon counts in the Gaussian approximation, from which we obtain the respective POVM. In Sec. III, we analyze the double homodyne scheme analogously to the previous section and show that the resulting POVM as projectors onto coherent states is not well defined for all asymmetry parameters, requiring generalization to the set of squeezed coherent states. In Sec. IV, we apply our results for homodyne and double homodyne detection to calculate the asymptotic secret fraction for the GG02 CV-QKD protocol. Finally, in Sec. V, we summarize the main results and outline directions for future research.

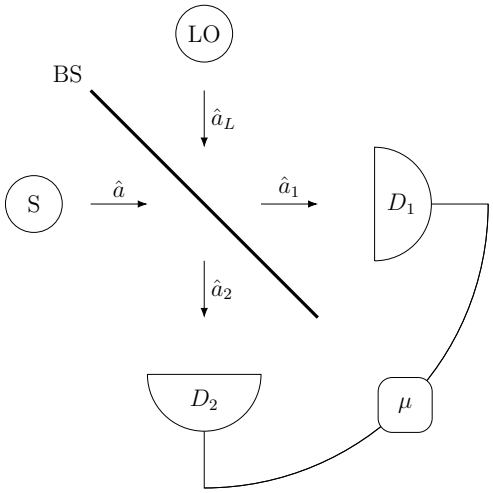


Figure 1: Scheme of a homodyne receiver: S is the source of the signal mode with the annihilation operator $\hat{a}$, LO is the source of the reference mode (local oscillator) with the annihilation operator $\hat{a}_L$, and BS is the beam splitter with the amplitude transmission and reflection coefficients $t = \cos\theta$ and $r = \sin\theta$, respectively; photodetectors D_1 and D_2 have quantum efficiencies η_1 and η_2, and $\mu \equiv m_1 - m_2$ is the photon count difference.

II. HOMODYNE DETECTION

A. Gaussian approximation

We begin with brief discussion of the homodyne measurement setup schematically depicted in Fig. 1. To this end, we assume that the beam spitter is unbalanced and its scattering matrix is chosen to be a real-valued rotation matrix with the transmission and reflection amplitudes, t and r, given by

$$t = \cos\theta \equiv C, \quad r = \sin\theta \equiv S. \tag{1}$$

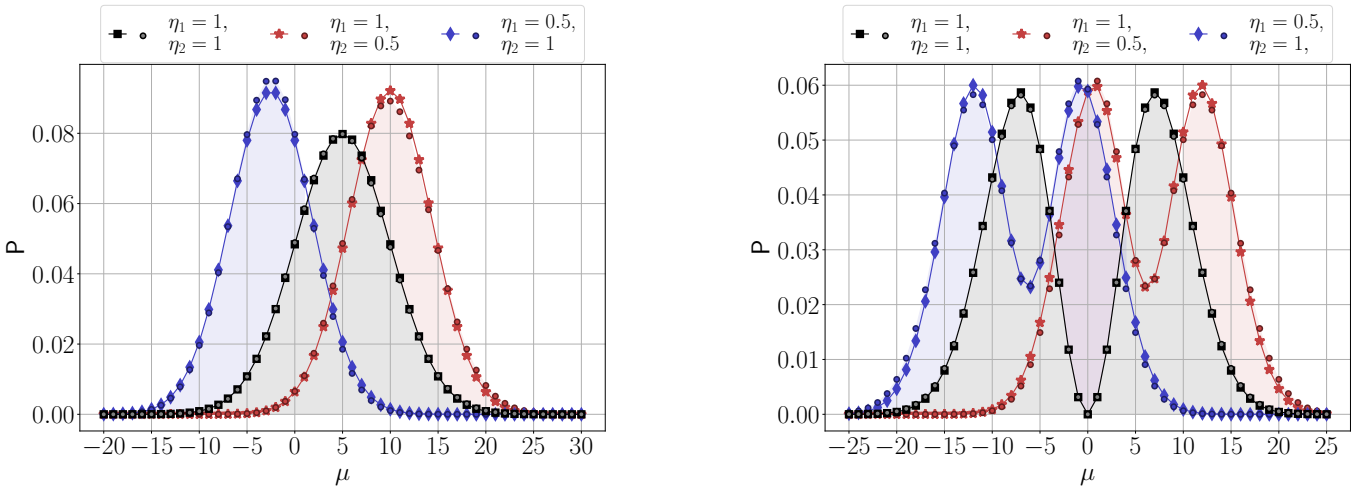
Then the input coherent states of the signal mode and the local oscillator are transformed into the output coherent states as follows

$$|\alpha, \alpha_L\rangle \mapsto |\alpha_1, \alpha_2\rangle, \tag{2}$$

$$\alpha_1 = C\alpha + S\alpha_L, \quad \alpha_2 = -S\alpha + C\alpha_L, \tag{3}$$

so that the joint probability of m_1 and m_2 photon counts for the photodetectors D_1 and D_2 can be computed from the well-known Kelley-Kleiner formula [8] (see also Ref. [7]):

$$\begin{aligned}\mathsf{P}(m_1, m_2) &= \langle \alpha_1, \alpha_2 | : \prod_{l=1}^{2} \frac{(\eta_l \hat{n}_l)^{m_l} e^{-\eta_l \hat{n}_l}}{m_l!} : |\alpha_1, \alpha_2\rangle \\ &= \prod_{l=1}^{2} \frac{(\eta_l |\alpha_L|^2)^{m_l}}{m_l!} e^{-\eta_l |\alpha_L|^2}\end{aligned} \tag{4}$$



(a) $|\psi\rangle = |\alpha\rangle$, $\alpha = 0.5$ (b) $|\psi\rangle = |n\rangle$, $n = 1$

Figure 2: Exact (circle dots) and approximate (solid lines with markers) statistical distributions of photon count difference for the signal mode prepared in (a) the coherent state and in (b) the single photon Fock state computed for for different efficiencies at $|\alpha_L| = 5$ and balanced beamspliiter.

where $:\ldots:$ stands for normal ordering, index $l \in \{1, 2\}$ labels output ports of the beam splitter, $\hat{n}_l = \hat{a}_l^\dagger \hat{a}_l$ is the photon number operator, m_l is the number of photon counts, η_l is the quantum efficiency of the detector D_l. We can now introduce the photon count difference

$$\mu = m_1 - m_2 \tag{5}$$

so that its statistical distribution can be written in the form of a product of the two Poisson distributions as follows

$$\mathsf{P}(\mu) = \sum_{m_2=\max(0,-\mu)}^{\infty} \frac{(\eta_1|\alpha_1|^2)^{\mu+m_2}}{(\mu+m_2)!} e^{-\eta_1|\alpha_1|^2} \times \frac{(\eta_2|\alpha_2|^2)^{m_2}}{m_2!} e^{-\eta_2|\alpha_2|^2}. \tag{6}$$

It is well known that, by performing summation over m_2, the probabilty $\mathsf{P}(\mu)$ reduces to the Skellam distribution given by [32]

$$\mathsf{P}(\mu) = e^{-\eta_1|\alpha_1|^2} e^{-\eta_2|\alpha_2|^2} \left(\frac{\eta_1|\alpha_1|^2}{\eta_2|\alpha_2|^2}\right)^{\mu/2} \times I_\mu\left(2\sqrt{\eta_1\eta_2|\alpha_1|^2|\alpha_2|^2}\right), \tag{7}$$

where $I_k(z)$ is the modified Bessel function of the first kind [33].

An important point is that, at sufficiently large $|\alpha_1|$ and $|\alpha_2|$, Poisson distributions that enter Eq. (4) can be approximated using the probability density functions of the normal distributions with mean and variance both equal to the mean of the corresponding Poisson distribution, $\lambda_i = \eta_i|\alpha_i|^2$. Then, in the continuum limit where summation in Eq. (6) is replaced with integration, the Skellam distribution (7) can be approximated assuming that the amplitude of the local oscillator, $|\alpha_L|$, is large (the strong-LO approximation) and we can apply the convolution formula for Gaussian probability densities

$$\int G(x_1 - x_2; \sigma_1) G(x_2; \sigma_2) \mathrm{d}x_2 = G(x_1; \sigma_1 + \sigma_2), \tag{8}$$

$$G(x; \sigma) \equiv \frac{1}{\sqrt{2\pi\sigma}} \exp\left(-\frac{x^2}{2\sigma}\right). \tag{9}$$

The above procedure immediately leads to the Gaussian approximation of the form:

$$\mathsf{P}_G(\mu) = G(\mu - \mu_G; \sigma_G), \tag{10}$$

$$\sigma_G = \eta_1|\alpha_1|^2 + \eta_2|\alpha_2|^2 \approx (\eta_1 S^2 + \eta_2 C^2)\,|\alpha_L|^2\,, \tag{11}$$

$$\mu_G = \eta_1|\alpha_1|^2 - \eta_2|\alpha_2|^2 \approx (\eta_1 S^2 - \eta_2 C^2)\,|\alpha_L|^2 + CS(\eta_1 + \eta_2)\,|\alpha_L|\,\langle\hat{x}_\phi\rangle \tag{12}$$

where

$$\langle\hat{x}_\phi\rangle \equiv \langle\alpha|\hat{x}_\phi|\alpha\rangle = 2\,\mathrm{Re}\,\alpha e^{-i\phi}, \quad \phi = \arg\alpha_L \tag{13}$$

is the average of the phase-rotated quadrature operator of the signal mode given by

$$\hat{x}_\phi = \hat{a}e^{-i\phi} + \hat{a}^\dagger e^{i\phi}. \tag{14}$$

Alternatively, the probability (10) can be rewritten in the form

$$\mathsf{P}_G(x) \equiv \mathsf{P}_G^{(\mathrm{H})}(x) = \frac{1}{\sqrt{2\pi\sigma_G}} \exp\left[-\frac{(x - \langle\hat{x}_\phi\rangle)^2}{2\sigma_x}\right], \tag{15}$$

where x is the quadrature variable given by

$$x \equiv \frac{\mu}{(\eta_1+\eta_2)CS\,|\alpha_L|} - \frac{\eta_1 S^2 - \eta_2 C^2}{(\eta_1+\eta_2)\,CS}\,|\alpha_L| \tag{16}$$

and σ_x is the quadrature variance

$$\sigma_x \equiv \frac{\eta_1 S^2 + \eta_2 C^2}{\left[(\eta_1+\eta_2)CS\right]^2}. \tag{17}$$

Minimizing the variance (17) with respect to the transmittance C^2 is straightforward:

$$\min_{C^2} \sigma_x = \left(\frac{\sqrt{\eta_1}+\sqrt{\eta_2}}{\eta_1+\eta_2}\right)^2 \geq 1. \tag{18}$$

B. Positive operator-valued measure

Our next step is to construct the positive operator-valued measure (POVM) based on the Gaussian approximation P_G. To this end, we note that the probability (15) is the expectation value of the POVM in the coherent state given by

$$\mathsf{P}_G = \langle\alpha|\hat{\Pi}_G|\alpha\rangle. \tag{19}$$

The operator $\hat{\Pi}_G$ can be regarded as a non-normalized density matrix describing a Gaussian state [30]. In the single mode case, such state is generally characterized by the 2×2 covariance matrix Σ_m and the vector of averaged quadratures $\mathbf{r}_m$. So, the Q-symbol of Π_G is determined by the Gaussian shaped distribution of the form:

$$\langle\alpha|\hat{\Pi}_G|\alpha\rangle \propto \exp\{-2(\mathbf{r}-\mathbf{r}_m)^{\mathrm{T}} \times (\Sigma_m + \mathbb{1})^{-1}(\mathbf{r}-\mathbf{r}_m)\}, \tag{20}$$

$$\alpha = \alpha_1 + i\alpha_2, \quad \mathbf{r} = \begin{pmatrix}\alpha_1\\ \alpha_2\end{pmatrix}, \tag{21}$$

where $\mathbb{1}$ is the identity matrix and the superscipt T stands for transposition, so that the transpose of $\mathbf{r}$ is the row vector: $\mathbf{r}^{\mathrm{T}} = (\alpha_1, \alpha_2)$.

It is known that POVMs of noisy measurements can be introduced using the dual of a quantum channel acting on POVMs representing perfect (noiseless) measurements [30]. In what follows we shall use the classical mixing (additive noise) quantum channel

$$\Phi_N : \hat{\Pi}_{\mathrm{id}} \mapsto \hat{\Pi}_G = \Phi_N(\hat{\Pi}_{\mathrm{id}}) = \frac{2}{\pi\sqrt{\det\Sigma_N}} \times \int_{\mathbb{C}} \mathrm{d}^2\beta \exp\{-2\mathbf{s}^T\Sigma_N^{-1}\mathbf{s}\}\hat{D}(\beta)\hat{\Pi}_{\mathrm{id}}\hat{D}(-\beta), \tag{22}$$

where $\mathbf{s}^{\mathrm{T}} = (\mathrm{Re}(\beta), \mathrm{Im}(\beta)) = (\beta_1, \beta_2)$, $\mathrm{d}^2\beta = \mathrm{d}\beta_1\mathrm{d}\beta_2$ and $\hat{D}(\beta)$ is the displacement operator given by

$$\hat{D}(\beta) = e^{\beta\hat{a}^\dagger - \beta^*\hat{a}}. \tag{23}$$

This is a self-dual channel, $\Phi_N^* = \Phi_N$, that transforms the POVM of the ideal projective Gaussian measurements, $\hat{\Pi}_{\mathrm{id}}$, into the POVM of the noisy measurements, $\hat{\Pi}_G$, by changing the covariance matrix Σ_{id} as follows

$$\Phi_N : \Sigma_{\mathrm{id}} \mapsto \Sigma_G = \Sigma_{\mathrm{id}} + \Sigma_N, \quad \Sigma_N \geq 0 \tag{24}$$

where Σ_N is the excess noise covariance matrix.

Our general strategy involves the following steps: (a) deriving the covariance matrix Σ_m from the Q-symbol of POVM; (b) identifying the states representing ideal measurements and Σ_{id}; and (c) constructing the POVM of the noisy measurements using the additive noise channel with the excess noise covariance matrix $\Sigma_N = \Sigma_m - \Sigma_{\mathrm{id}}$.

In the case of the perfectly symmetric noiseless homodyne measurement with $\eta_1 = \eta_2 = 1$ and $C = S = 1/\sqrt{2}$, the average (19) takes the form

$$\mathsf{P}_{\mathrm{id}}^{(\mathrm{H})} = \frac{1}{|\alpha_L|} Q_{x,\phi}(\alpha), \tag{25}$$

where $x = \mu/|\alpha_L|$ and $Q_{x,\phi}(\alpha)$ is the Husimi Q distribution for the eigenstate of the phase-rotated quadrature operator (14), $|x,\phi\rangle$, given by (see, e.g., the textbook [9])

$$Q_{x,\phi}(\alpha) = |\langle\alpha|x,\phi\rangle|^2 = G(x - \langle\hat{x}\rangle_\phi; 1). \tag{26}$$

Thus, we are led to the well-known result that, in the Gaussian approximation, the POVM describing sharp homodyne measurements

$$\hat{\Pi}_{\mathrm{id}}^{(\mathrm{H})} = \frac{1}{|\alpha_L|}|x,\phi\rangle\langle x,\phi| \tag{27}$$

is proportional to the projector onto $|x,\phi\rangle$.

Since the eigenstate of the rotated quadrature $\hat{x}_\phi$ can be represented as the squeezed coherent state taken in the limit of infinitely large squeezing

$$|x,\phi\rangle = \hat{R}(\phi)|x\rangle, \quad \hat{R}(\phi) = \exp\{i\phi\hat{a}^\dagger\hat{a}\}, \tag{28}$$

$$|x\rangle = \lim_{r\to+\infty}\sqrt{\frac{\cosh(r)}{(2\pi)^{1/2}}}\hat{D}(x/2)\hat{S}(-r)|0\rangle, \tag{29}$$

where $\hat{S}(r)$ is the squeezing operator given by

$$\hat{S}(\zeta) = e^{\frac{1}{2}\left(\zeta\hat{a}^{\dagger 2} - \zeta^*\hat{a}^2\right)}, \tag{30}$$

its covariance matrix assumes similar form [30]:

$$\Sigma_{\mathrm{id}}^{(\mathrm{H})} = \lim_{r\to+\infty} R(\phi)\,\mathrm{diag}(e^{-2r}, e^{2r})\,R^{\mathrm{T}}(\phi), \tag{31}$$

where R_ϕ is the rotation matrix given by

$$R(\phi) = \begin{pmatrix}\cos\phi & -\sin\phi\\ \sin\phi & \cos\phi\end{pmatrix}. \tag{32}$$

In a more general asymmetric case with $\eta_1 \neq \eta_2$ and $C \neq S$, from the Q symbol (15) we deduce the following expression for the covariance matrix of noisy homodyne measurements

$$\Sigma_m^{(\mathrm{H})} = \lim_{r\to+\infty} R(\phi)\,\mathrm{diag}(e^{-2r}+\sigma_N, e^{2r})\,R^{\mathrm{T}}(\phi) = \Sigma_{\mathrm{id}}^{(\mathrm{H})} + \Sigma_N^{(\mathrm{H})}, \quad \Sigma_N^{(\mathrm{H})} = R(\phi)\,\mathrm{diag}(\sigma_N, 0)\,R^{\mathrm{T}}(\phi), \tag{33}$$

where σ_N is the excess noise given by

$$\sigma_N = \sigma_x - 1 \geq 0. \tag{34}$$

Note that the excess noise variance σ_N takes into account asymmetry effects and its non-negativity is guarantied by inequality (18).

We can now use formula (22) along with the algebraic identities

$$\int_{\mathbb{C}} \mathrm{d}^2\beta \exp\{-2\mathbf{s}^T\big[R(\phi)\Sigma_N^{-1}R^{\mathrm{T}}(\phi)\big]\mathbf{s}\}\hat{D}(\beta)\hat{\Pi}_{\mathrm{id}}\hat{D}(-\beta)$$
$$= \int_{\mathbb{C}} \mathrm{d}^2\beta \exp\{-2\mathbf{s}^T\Sigma_N^{-1}\mathbf{s}\}\hat{D}(\beta e^{i\phi})\hat{\Pi}_{\mathrm{id}}\hat{D}(-\beta e^{i\phi}), \tag{35}$$
$$\hat{D}(\beta e^{i\phi})\hat{R}(\phi)\hat{D}(\alpha_m) = \hat{R}(\phi)\hat{D}(\beta)\hat{D}(\alpha_m) \tag{36}$$

to derive the POVM in the form of randomly displaced projectors

$$\hat{\Pi}_G^{(\mathrm{H})} = \frac{1}{(\eta_1+\eta_2)CS\,|\alpha_L|} \times \int \mathrm{d}x' G\,(x-x';\sigma_N)\,|x',\phi\rangle\langle x',\phi|. \tag{37}$$

Alternatively, this result can be obtained using the convolution identity (8) that leads to the relation

$$\mathsf{P}_G^{(\mathrm{H})}(x) = \sqrt{\frac{\sigma_x}{\sigma_G}} \int \mathrm{d}x'\, G(x-x';\sigma_N) Q_{x',\phi}(\alpha) \tag{38}$$

linking the Q symbols of the projectors $|x',\phi\rangle\langle x',\phi|$ and the POVM $\hat{\Pi}_G^{(\mathrm{H})}$.

The exact and approximate analytical results for photon count difference statistical distributions, given by Eq. (7) and Eq. (10) respectively, are valid for the case where the LO and signal modes are both in the coherent states. In the more general case when the quantum state of the signal mode is $|\psi\rangle$, the probability distributions can be evaluated using the relations

$$\mathsf{P}(\mu;|\psi\rangle) = \int P_{|\psi\rangle}(\alpha)\mathsf{P}(\mu;\alpha)\mathrm{d}^2\alpha,$$
$$\mathsf{P}_G(x;|\psi\rangle) = \langle\psi|\hat{\Pi}_G|\psi\rangle, \tag{39}$$

where $P_{|\psi\rangle}(\alpha)$ is the Glauber P function of the quantum state $|\psi\rangle$. In Fig. 2, we show the results computed for the single-photon Fock state obtained utilizing the well-known expression for the P-function of Fock states $|\psi\rangle = |n\rangle$ given by

$$P_{|n\rangle}(\alpha) = \frac{e^{|\alpha|^2}}{n!}\Big(\frac{\partial^2}{\partial\alpha\partial\alpha^*}\Big)^n \delta^2(\alpha), \tag{40}$$

where $\delta^2(\alpha) = \delta(\mathrm{Re}\,\alpha)\delta(\mathrm{Im}\,\alpha)$.

Figure 2 displays the photocount difference probabilities computed from the exact and Gaussian probability distributions for the balanced beam splitter at different photodetector efficiencies and signal mode input states.

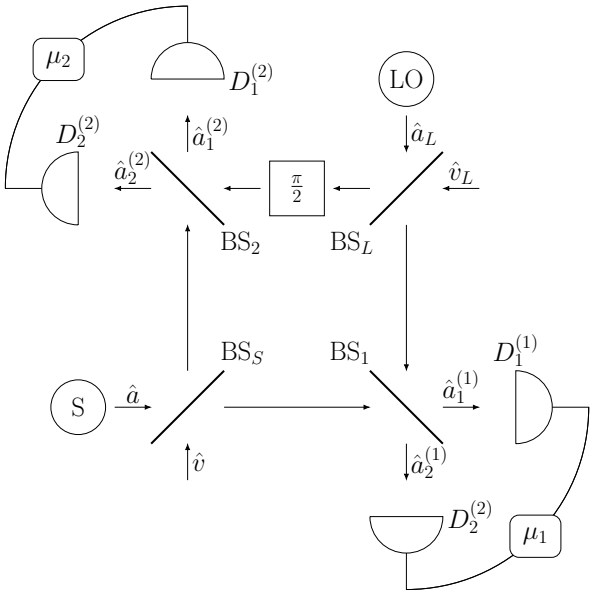


Figure 3: Scheme of an eight port double homodyne receiver. S is the source of the signal mode $\hat{a}$; LO is the source of the reference mode $\hat{a}_L$; BS_S (BS_L) is the signal mode (local oscillator) beam splitter; $\frac{\pi}{2}$ is the quarter wave phase shifter; BS_i is the beam splitter of ith homodyne; $D_{1,2}^{(i)}$ are the photodetectors of the ith homodyne; and $\mu_i = m_1^{(i)} - m_2^{(i)}$ is the photon count difference registered by the detectors of ith homodyne.

Fig. 2a shows that, in agreement with Eq. (12), asymmetry in photodetection results in the shift of the probability maximum.

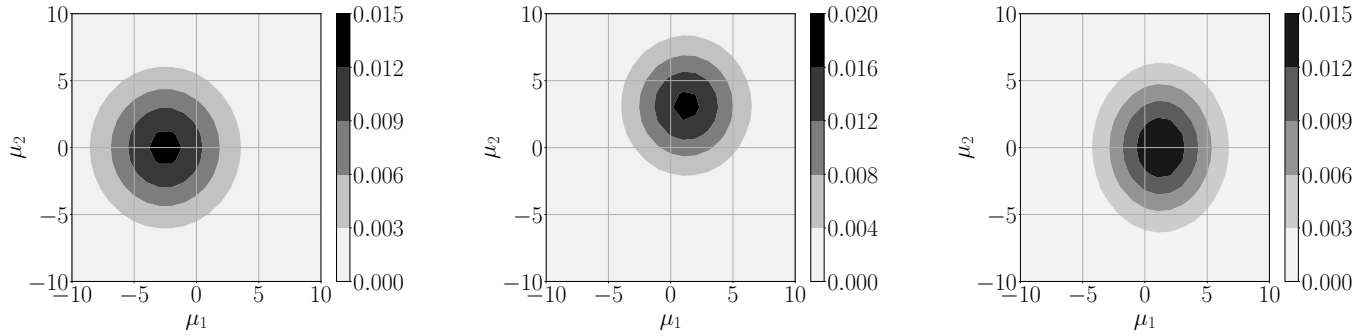
The distributions for the single-photon states are depicted in Fig. 2b and, similar to the coherent state, demonstrate the effect of asymmetry-induced shift. Another noticeable effect is that the probability minimum between the central and side peaks become less pronounced.

From Fig. 2 it becomes apparent that perfomance of Gaussian approximation worsens in presence of asymmetry, which we will quantify in Appendix A.

Our concluding remark concerns an alternative method to approximate Eq. (7) with a Gaussian-shaped distribution which is based on the asymptotic expansions of the modified Bessel functions. In Appendix B we show that, for the asymmetric homodyne scheme, this method generally leads to ill-posed POVMs because the corresponding quadrature variance appears to be too small leading to negative contribution of the excess noise.

III. DOUBLE HOMODYNE DETECTION

In this section, we consider the eight-port double homodyne scheme depicted in Fig. 3 (see, e.g., Refs. [9, 34]). This measurement scheme is known to allow reconstructing the Husimi Q-function of the signal state [5] pro-



(a) $\eta_1^{(i)} = 1, \eta_2^{(i)} = 1$ (b) $\eta_1^{(i)} = 1, \eta_2^{(i)} = 0.5$ (c) $\eta_1^{(i)} = 1, \eta_2^{(1)} = 1, \eta_2^{(2)} = 0.5$

Figure 4: Double homodyne statistical distribution of photocount differences computed from Eq. (42) for various detector efficiencies at $\alpha = 0.5$ and $\alpha_L = 5$. All the beam splitters are taken to be balanced.

viding complete information about the signal state that may be used in CV-QKD protocols. It should be noted that restoration of the complex amplitude of the state completely serves as the basis for composable security proofs [1, 6].

Figure 3 shows two homodyne setups with elements such as beam splitters BS_i and photodetectors $D_{1,2}^{(i)}$ labeled by the index $i \in \{1, 2\}$. Referring to Fig. 3, the LO and signal modes are transmitted through the beam splitters BS_L and BS_S, respectively. Similar to the analysis performed in the previous section, we assume that the modes are in the coherent states, $|\alpha\rangle$ and $|\alpha_L\rangle$. So, we have the amplitudes

$$\alpha^{(1)} = C_S\alpha, \quad \alpha^{(2)} = S_S\alpha,$$
$$\alpha_L^{(1)} = C_L\alpha_L, \quad \alpha_L^{(2)} = -iS_L\alpha_L, \tag{41}$$

where $\alpha^{(i)}$ ($\alpha_L^{(i)}$) stands for the amplitude describing the input coherent state of the signal (LO) mode of the homodyne labeled by the upper index $i \in \{1, 2\}$. Note that the phase factor $-i = e^{-i\pi/2}$ in the expression for $\alpha_L^{(2)}$ is introduced by a suitably chosen phase shifter placed before the corresponding input port of the beam splitter BS_2.

Direct calculation shows that the joint statistics of the difference photocount events is determined by the product of the Skellam distributions given by

$$\mathsf{P}(\mu_1, \mu_2) = \mathsf{P}_1(\mu_1)\mathsf{P}_2(\mu_2), \quad \mu_i = m_1^{(i)} - m_2^{(i)}, \tag{42}$$
$$\mathsf{P}_i(\mu_i) = e^{-\eta_1^{(i)}|\alpha_1^{(i)}|^2} e^{-\eta_2^{(i)}|\alpha_2^{(i)}|^2} \left(\frac{\eta_1^{(i)}|\alpha_1^{(i)}|^2}{\eta_2^{(i)}|\alpha_2^{(i)}|^2}\right)^{\mu_i/2}$$
$$\times I_{\mu_i}\big(2\sqrt{\eta_1^{(i)}\eta_2^{(i)}|\alpha_1^{(i)}|^2|\alpha_2^{(i)}|^2}\big), \tag{43}$$

where, similar to the homodyne scheme, the amplitudes of the coherent states at the output ports of the beam splitter BS_i

$$\alpha_1^{(i)} = C_i\alpha^{(i)} + S_i\alpha_L^{(i)}, \quad \alpha_2^{(i)} = -S_i\alpha^{(i)} + C_i\alpha_L^{(i)} \tag{44}$$

are expressed in terms of the transmission and reflection amplitudes, $C_i = \cos\theta_i$ and $S_i = \sin\theta_i$.

We can now apply Eqs. (10)-(12) to approximate each Skellam distribution on the right hand side of Eq. (42) and derive the Gaussian approximation for the double homodyne scheme in the form:

$$\mathsf{P}_G(\mu_1, \mu_2) = G(\mu_1 - \mu_G^{(1)}; \sigma_G^{(1)})G(\mu_2 - \mu_G^{(2)}; \sigma_G^{(2)}), \tag{45}$$
$$\sigma_G^{(i)} = (\eta_1^{(i)}S_i^2 + \eta_2^{(i)}C_i^2)|\alpha_L^{(i)}|^2, \tag{46}$$
$$\mu_G^{(i)} = (\eta_1^{(i)}S_i^2 - \eta_2^{(i)}C_i^2)|\alpha_L^{(i)}|^2 + C_iS_i(\eta_1^{(i)} + \eta_2^{(i)})|\alpha_L^{(i)}|$$
$$\times 2\,\mathrm{Re}\,\alpha^{(i)}e^{-i\phi}, \quad \phi = \arg\alpha_L. \tag{47}$$

Similar to Eq. (15), it is useful to put the probability (45) into the folowing quadrature form:

$$\mathsf{P}_G(x_1, x_2) \equiv \mathsf{P}_G^{(\mathrm{DH})}(x_1, x_2) = \frac{1}{2\pi\sqrt{\sigma_G^{(1)}\sigma_G^{(2)}}}$$
$$\times \exp\left[-\frac{(x_1 - \mathrm{Re}\,\alpha e^{-i\phi})^2}{\sigma_1} - \frac{(x_2 - \mathrm{Im}\,\alpha e^{-i\phi})^2}{\sigma_2}\right] \tag{48}$$

where x_i are the quadrature variables given by

$$x_1 = \frac{1}{2(\eta_1^{(1)} + \eta_2^{(1)})C_1S_1C_S}$$
$$\times \left\{\frac{\mu_1}{|\alpha_L^{(1)}|} - (\eta_1^{(1)}S_1^2 - \eta_2^{(1)}C_1^2)|\alpha_L^{(1)}|\right\}, \tag{49}$$
$$x_2 = \frac{1}{2(\eta_1^{(2)} + \eta_2^{(2)})C_2S_2S_S}$$
$$\times \left\{\frac{\mu_2}{|\alpha_L^{(2)}|} - (\eta_1^{(2)}S_2^2 - \eta_2^{(2)}C_2^2)|\alpha_L^{(2)}|\right\}, \tag{50}$$

and relations

$$\sigma_1 = \frac{\sigma_x^{(1)}}{2C_S^2}, \quad \sigma_2 = \frac{\sigma_x^{(2)}}{2S_S^2}, \tag{51}$$
$$\sigma_x^{(i)} = \frac{\eta_1^{(i)}S_i^2 + \eta_2^{(i)}C_i^2}{\left[(\eta_1^{(i)} + \eta_2^{(i)})C_iS_i\right]^2} \tag{52}$$

give the quadrature variances σ_1 and σ_2.

As in Sec. II, formula (48) giving the Q-symbol of POVM (see Eq. (19)) provides the starting point for reconstruction of the POVM describing the double homodyne measurements. The covariance matrix of the POVM reads

$$\Sigma_m^{(\mathrm{DH})} = R(\phi)\,\mathrm{diag}(\delta_1,\delta_2)R^{\mathrm{T}}(\phi), \tag{53}$$
$$\delta_i = 2\sigma_i - 1, \tag{54}$$

whereas the column vector of the averaged quadratures is given by

$$\mathbf{r}_m^{(\mathrm{DH})} = R(\phi)\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}. \tag{55}$$

In the case, where all the beam splitters are balanced and the photodetection is perfect, we have

$$\mathsf{P}_{\mathrm{CS}}^{(\mathrm{DH})}(x_1,x_2) = \frac{\left|\langle\alpha|ze^{i\phi}\rangle\right|^2}{\pi\left|\alpha_L\right|^2},$$
$$\left|\langle\alpha|ze^{i\phi}\rangle\right|^2 = e^{-|ze^{i\phi}-\alpha|^2}, \quad z = x_1 + ix_2, \tag{56}$$

so that the POVM is proportional to the projector onto the coherent state $\hat{R}(\phi)|z\rangle = |(x_1+ix_2)\,e^{i\phi}\rangle$:

$$\hat{\Pi}_{\mathrm{CS}}^{(\mathrm{DH})}(x_1,x_2) = \frac{1}{\pi\left|\alpha_L\right|^2}|ze^{i\phi}\rangle\langle ze^{i\phi}|, \tag{57}$$

with the coherent state covariance matrix

$$\Sigma_{\mathrm{CS}} = \mathbb{1} \tag{58}$$

equal to the identity matrix.

For nonideal measurements, following the steps of the general procedure described in Sec. II B, we need to identify the projectors representing the ideal measurements. At this stage, it is natural to assume these projectors are given by the coherent state projectors $|ze^{i\phi}\rangle\langle ze^{i\phi}|$. So, the covariance matrix of the additive noise quantum channel (22) is given by

$$\tilde{\Sigma}_N^{(\mathrm{DH})} = \Sigma_m^{(\mathrm{DH})} - \Sigma_{\mathrm{CS}}$$
$$= 4R(\phi)\,\mathrm{diag}(\tilde{\sigma}_N^{(1)},\tilde{\sigma}_N^{(2)})R^{\mathrm{T}}(\phi), \tag{59}$$
$$4\tilde{\sigma}_N^{(i)} = \delta_i - 1 = 2(\sigma_i - 1). \tag{60}$$

It is not difficult to check with the help of the convolution relation (8) that the probability (48) can be expressed as a Gaussian superposition of the coherent state Husimi functions

$$\mathsf{P}_G^{(\mathrm{DH})}(x_1,x_2) = \frac{\sqrt{\sigma_1\sigma_2}}{2\pi\sqrt{\sigma_G^{(1)}\sigma_G^{(2)}}}\int \mathrm{d}^2\beta\, G(x_1-\beta_1;\tilde{\sigma}_N^{(1)})$$
$$\times\, G(x_2-\beta_2;\tilde{\sigma}_N^{(2)})|\langle\beta e^{i\phi}|\alpha\rangle|^2, \tag{61}$$

where $\beta = \beta_1 + i\beta_2$, and $\sigma_N^{(i)}$ is the excess noise variance given by Eq. (60). The corresponding expression for the POVM reads

$$\hat{\Pi}_G^{(\mathrm{DH})}(x_1,x_2) = \frac{\sqrt{\sigma_1\sigma_2}}{2\pi\sqrt{\sigma_G^{(1)}\sigma_G^{(2)}}}\int \mathrm{d}^2\beta\, G(x_1-\beta_1;\tilde{\sigma}_N^{(1)})$$
$$\times\, G(x_2-\beta_2;\tilde{\sigma}_N^{(2)})|\beta e^{i\phi}\rangle\langle\beta e^{i\phi}|. \tag{62}$$

An important point is that the results given by Eq. (61) and Eq. (62) are well defined only if the excess noise covariance matrix (59) is positive semi-definite: $\tilde{\Sigma}_N^{(\mathrm{DH})} \geq 0$. The latter requires the variances $\tilde{\sigma}_N^{(2)}$ and $\tilde{\sigma}_N^{(2)}$ to be nonnegative, which is equivalent to the inequalities: $\sigma_1 \geq 1$ and $\sigma_2 \geq 1$ (see Eq. (60)). From Eq. (51), we arrive at the conditions $\sigma_x^{(1)} \geq 2C_S^2$ and $\sigma_x^{(2)} \geq 2S_S^2$ be met.

When the beam splitter BS_S is balanced $2C_S^2 = 2S_S^2 = 1$, applicability of the above expression for the POVM is guaranteed by the inequality (see Eq. (18))

$$\sigma_x^{(i)} \geq \left(\frac{\sqrt{\eta_1^{(i)}} + \sqrt{\eta_2^{(i)}}}{\eta_1^{(i)} + \eta_2^{(i)}}\right)^2 \geq 1. \tag{63}$$

Otherwise, either $2C_S^2$ or $2S_S^2$ will be above unity, and the coherent-state representation is valid only if the value of the corresponding variance $\sigma_x^{(i)}$ is sufficiently high. For example, at $\eta_{1,2}^{(i)} = \eta < 1/2$, the minimal values of $\sigma_x^{(i)}$ are higher than 2 (see Eq. (63)) and the noise variance will be positive at any disbalance of the signal mode beam splitter because $\max\{2C_S^2, 2S_S^2\} \leq 2$.

When one of the noise variances (60) is negative, the POVM reconstruction procedure needs to be generalized. We shall present details on this generalization in the next section. Meanwhile, in the concluding part of this section, we confine ourselves to the cases where $\sigma_N^{(i)}$ are positive.

The effects of photodetection asymmetry are illustrated in Fig. 4 which presents numerical results for the double homodyne distribution (42) in the photocont difference μ_1-μ_2 plane. Referring to Fig. 4, in addition to the shift of the distribution, the asymmetry induced difference of the variances at $\eta_1^{(1)} + \eta_2^{(1)} \neq \eta_1^{(2)} + \eta_1^{(2)}$ manifests itself as the two dimensional anisotropy of the double homodyne distribution.

A. Positive operator-valued measure and squeezed states

In this section we apply our general procedure outlined in Sec. II B to double homodyne measurements so as to derive the POVM expressed in terms of the projectors describing sharp (ideal) measurements. According to the previous section, the class of such measurement states, which are assumed to be Gaussian, should be extended beyond the family of the coherent states even though these states represent the measurements in the limiting case of balanced beam splitters and perfect photodetection. So, we begin with the generalized representation

of noiseless measurements that uses a set of pure states enlarged so as to include the squeezed states taken in the form

$$|z,r,\phi\rangle = \hat{R}(\phi)\hat{D}(z)\hat{S}(r)|0\rangle = \hat{D}(ze^{i\phi})\hat{S}(re^{2i\phi})|0\rangle, \quad (64)$$

where $\hat{D}(\beta)$ ($\hat{S}(\zeta)$) is the displacement (squeezing) operator given by Eq. (23) (Eq. (30)).

Then the covariance matrix of the noiseless POVM proporional to the projectors $|z,r,\phi\rangle\langle z,r,\phi|$ is given by

$$\Sigma_{\rm id}^{\rm (DH)} = R(\phi)\begin{pmatrix} e^{2r} & 0 \\ 0 & e^{-2r}\end{pmatrix} R^{\rm T}(\phi), \quad (65)$$

whereas formula (55) giving the vector of the averaged quadratures remains applicable to the case of the squeezed states As a result, the non-normalized Husimi distribution for the squeezed state (64) takes the form

$$\left|\langle ze^{i\phi}, re^{2i\phi}|\alpha\rangle\right|^2 = \frac{1}{\cosh r}\exp\left[-\frac{2}{e^{2r}+1}\left(x_1-\tilde{\alpha}_1\right)^2\right.$$
$$\left. -\frac{2}{e^{-2r}+1}\left(x_2-\tilde{\alpha}_2\right)^2\right], \quad (66)$$

$$\tilde{\alpha}_1 = {\rm Re}\,\alpha e^{-i\phi}, \quad \tilde{\alpha}_2 = {\rm Im}\,\alpha e^{-i\phi}. \quad (67)$$

From Eqs. (53) and (65), we derive the covariance matrix of the additive noise channel

$$\Sigma_N^{\rm (DH)}(r) = \Sigma_m^{\rm (DH)} - \Sigma_{\rm id}^{\rm (DH)}$$
$$= R(\phi)\begin{pmatrix} 4\sigma_N^{(1)}(r) & 0 \\ 0 & 4\sigma_N^{(2)}(r)\end{pmatrix} R^{\rm T}(\phi) \geq 0, \quad (68)$$

where

$$4\sigma_N^{(1)}(r) = \delta_1 - e^{2r}, \quad 4\sigma_N^{(2)}(r) = \delta_2 - e^{-2r}. \quad (69)$$

Since this matrix should be positive semidefinite, the noise variances (69) cannot take negative values.

For subsequent analysis, it will be useful to express the covariance matrix elements δ_i given by Eq. (54) in terms of the disbalance (reflection-to-transmission) ratio of the input beam splitter, q, and the parameters $\sigma_x^{(1)}$ and $\sigma_x^{(2)}$ (see Eq. (52)):

$$\delta_1 \equiv 2\sigma_1 - 1 = (q+1)\sigma_x^{(1)} - 1 \geq q = \frac{S_S^2}{C_S^2}, \quad (70a)$$

$$\delta_2 \equiv 2\sigma_2 - 1 = (q^{-1}+1)\sigma_x^{(2)} - 1 \geq q^{-1} = \frac{C_S^2}{S_S^2}, \quad (70b)$$

where we used the fact that $\sigma_x^{(i)} \geq 1$ (see Eq. (63)).

We assume without the loss of generality that the reflectance of the input beam splitter ${\rm BS}_S$ is larger than its transmittance, so that the disbalance ratio is above unity

$$q = \frac{1-C_S^2}{C_S^2} = \frac{S_S^2}{C_S^2} \geq 1. \quad (71)$$

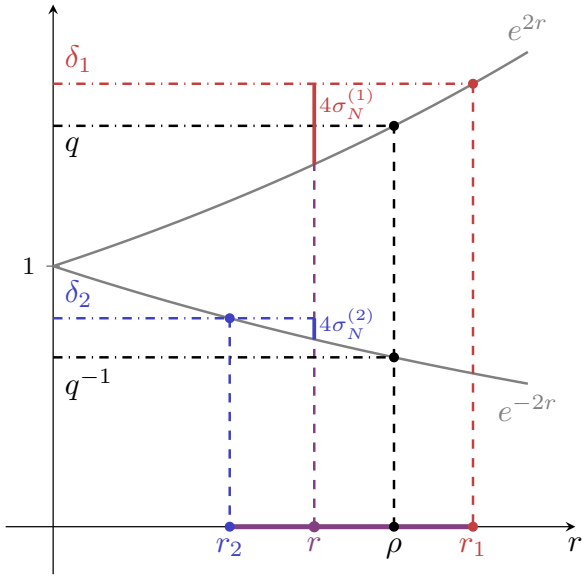


Figure 5: Geometry in the δ-r (principal variance-squeezing parameter) plane. Noise variances (69) are positive when the principal variance δ_1 (δ_2) is above e^{2r} (e^{-2r}), so that the squeezing parameter r is ranged between r_1 and r_2. Solid grey lines represent the graphs of the exponents: e^{2r} and e^{-2r}. In the case, where $\delta_1 = q$ and $\delta_2 = q^{-1}$, the interval is reduced to the point $r_1 = r_2 = \rho$.

As is shown in Fig. 5, this implies that δ_1 is above unity: $\delta_1 \geq q \geq 1$, whereas the minimal value of δ_2 is $q^{-1} \leq 1$. It is illustrated that the noise variances are positive provided the squeezing parameter r is ranged between the endpoints of the interval given by

$$r \in [r_2, r_1], \; r_1 = \ln \delta_1^{1/2}, \; r_2 = \ln \delta_2^{-1/2} \quad (72)$$

An important point is that the value of the squeezing parameter is generally not uniquely determined by the conditions (69). Referring to Fig. 5, the sole exception is provided by the limiting case of perfect homodyne measurements with $\sigma_x^{(1)} = \sigma_x^{(2)} = 1$ and $\delta_1 = \delta_2^{-1} = q$. In this case, the squeezing parameter is uniquely defined by the disbalance ration of the input beam splitter

$$r_1 = r_2 = \rho = \frac{1}{2}\ln q = \ln\sqrt{\frac{1-C_S^2}{C_S^2}} \quad (73)$$

and the probability (48) expressed in terms of the distribution (66)

$$\mathsf{P}_G^{(1)}(x_1,x_2) = \frac{\cosh\rho}{2\pi|\alpha_L^{(1)}\alpha_L^{(2)}|}\left|\langle ze^{i\phi}, \rho e^{2i\phi}|\alpha\rangle\right|^2, \quad (74)$$

yields the POVM

$$\hat{\Pi}_G^{(1)}(x_1,x_2) = \frac{\cosh\rho}{2\pi|\alpha_L^{(1)}\alpha_L^{(2)}|}$$
$$\times |ze^{i\phi}, \rho e^{2i\phi}\rangle\langle ze^{i\phi}, \rho e^{2i\phi}|, \quad (75)$$

which is propotional to the projector onto the pure squeezed state $|ze^{i\phi}, \rho e^{2i\phi}\rangle$. The result (73) was reported in Ref. [35].

When the homodyne measurements are not perfect due to unbalanced beam splitters and nonideal photodetectors, the squeezing parameter is no longer uniquely defined. So, in the interval (72), we have the decomposition of the probability (48)

$$\mathsf{P}_G^{(\mathrm{DH})}(x_1,x_2) = \frac{\sqrt{\sigma_1\sigma_2}}{2\pi\sqrt{\sigma_G^{(1)}\sigma_G^{(2)}}} \int \mathrm{d}^2\beta G(x_1-\beta_1;\sigma_N^{(1)}(r)) \times G(x_2-\beta_2;\sigma_N^{(2)}(r)) \left|\langle \beta e^{i\phi}, re^{2i\phi}|\alpha\rangle\right|^2, \tag{76}$$

that varies with the squeezing parameter r. Similarly, the corresponding POVM

$$\hat{\Pi}_G^{(\mathrm{DH})}(x_1,x_2) = \frac{\sqrt{\sigma_1\sigma_2}}{2\pi\sqrt{\sigma_G^{(1)}\sigma_G^{(2)}}} \int \mathrm{d}^2\beta G(x_1-\beta_1;\sigma_N^{(1)}(r)) \times G(x_2-\beta_2;\sigma_N^{(2)}(r)) |\beta e^{i\phi}, re^{2i\phi}\rangle\langle \beta e^{i\phi}, re^{2i\phi}|, \tag{77}$$

decomposed into the Gaussian incoherent superposition of the pure squeezed states explicitly depends on the value of r.

Our analysis suggests that the ambiguity (non-uniqueness) of the Gaussian represention (77) for the double-homodyne POVM is a universal feature coming into play in the presence of imperfections. Even when $\delta_2 \geq 1$ and the coherent-state represention (62) for the POVM is well-defined, the photocount statistics can be reproduced using the squeezed-state representation (77) with $r \neq 0$. Analogously, the density matrices of mixed Gaussian states can be expressed as randomly displaced squeezed states and, in general, this representation is non-unique. In the next section we demonstrate that the non-uniqueness of additive noise channel should be taken into account in certain scenarios of security analysis of continuous variable quantum key distribution protocols.

should be taken into account when analyzing

IV. MEASUREMENT IMPERFECTIONS AND SECURITY OF CONTINUOUS VARIABLE QUANTUM KEY DISTRIBUTION PROTOCOLS

In this section, we apply the developed formalism for asymmetric noisy Gaussian measurements to analyze the Gaussian-modulated coherent-state (GMCS, or GG02 [31]) continuous variable quantum key distribution (CV-QKD) protocol. Specifically, we evaluate the mutual information, Holevo information, and asymptotic secret fraction (secure key rate per symbol) under the untrusted-noise scenario to quantify how measurement asymmetry affects the protocol security.

In the GG02 protocol, the sender (Alice) prepares an ensemble of normally distributed coherent states $|\alpha = \alpha_1 + i\alpha_2\rangle$ with the Gaussian probability density $p_{\mathrm{A}}(\alpha) = G(\alpha_1;\frac{V_A}{4})G(\alpha_2;\frac{V_A}{4})$, so that the density matrix of the input state

$$\hat{\rho}_{\mathrm{A}} = \int \mathrm{d}^2\alpha\, p_{\mathrm{A}}(\alpha)|\alpha\rangle\langle\alpha| = \Phi_A(|0\rangle\langle 0|),$$
$$p_{\mathrm{A}}(\alpha) = \frac{2}{\pi V_{\mathrm{A}}}\exp\left[-\frac{2|\alpha|^2}{V_{\mathrm{A}}}\right] \tag{78}$$

represents randomly displaced states generated by applying the additive noise channel Φ_A with the covariance matrix $\Sigma_A = V_A\mathbb{1}$ to the vacuum state $|0\rangle\langle 0|$. The states prepared by Alice are then transmitted to the receiver (Bob) through the quantum channel $\mathcal{E}_{A\to B}$ which is assumed to be a composition of two Gaussian channels [36] given by

$$\mathcal{E}_{A\to B} = \Phi_\xi \circ \mathcal{E}_T, \tag{79}$$

where $\mathcal{E}_T$ is the pure-loss channel with the transmittance T and Φ_ξ is the additive noise channel with the noise covariance matrix $\Sigma_\xi = \xi\mathbb{1}$. In order to decode the signal, Bob measures the output state of the quantum channel (79). Joint probability distribution of the outcomes of measurements and the components of the coherent state amplitude determines the mutual information between Alice and Bob. When Bob performs homodyne detection, the mutual information is given by

$$I_{\mathrm{AB}}^{(\mathrm{H})} = \frac{1}{2}\log\left[1+\frac{TV_{\mathrm{A}}}{\sigma_x+\xi}\right], \tag{80}$$

where σ_x is given by Eq. (17). If Bob uses double-homodyne detection, the expression for the mutual information takes the form

$$I_{\mathrm{AB}}^{(\mathrm{DH})} = \frac{1}{2}\sum_i \log\left[1+\frac{TV_{\mathrm{A}}}{2\sigma_i+\xi}\right], \tag{81}$$

where σ_i are given by Eq. (52). The details on derivation of these expressions are relegated to Appendix C.

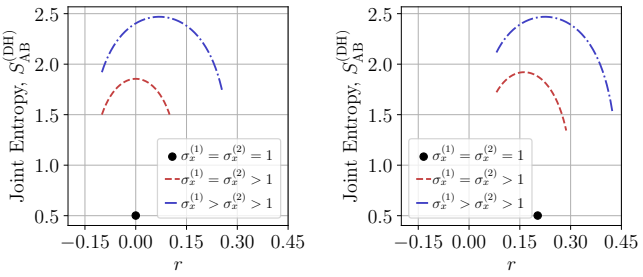
Using the equivalence between the prepare-and-measure (PM) and entanglement-based (EB) pictures (see Appendix C),

the Holevo information can be derived from the covariance matrix of the two mode squeezed vacuum (TMSV) state $\Sigma_{\mathrm{AB}}^{\mathrm{EB}}$. After transmitting through noisy channel (see Ref. [36]), TMSVSs covariance matrix reads

$$\begin{aligned}\mathcal{I}_A\otimes\Phi_\xi\circ\mathcal{E}_T:\quad &\Sigma_{\mathrm{TMSV}}\\ &= \begin{pmatrix} V\mathbb{1} & \sqrt{V^2-1}\sigma_z \\ \sqrt{V^2-1}\sigma_z & V\mathbb{1}\end{pmatrix} \mapsto \Sigma_{\mathrm{AB}}^{\mathrm{EB}}\\ &= \begin{pmatrix} V\mathbb{1} & c\sigma_z \\ c\sigma_z & V_{\mathrm{B}}\mathbb{1}\end{pmatrix}, \quad \sigma_z = \mathrm{diag}(1,-1),\end{aligned} \tag{82}$$

where

$$V = 1+V_{\mathrm{A}}, \quad c = \sqrt{T(V^2-1)}, \tag{83}$$
$$V_{\mathrm{B}} = T(V-1)+1+\xi. \tag{84}$$



(a) $q = 1$ (b) $q > 1$

Figure 6: Dependence of joint entropy on squeezing parameter r (See Eq. (72)) for different quadrature variances $\sigma_x^{(i)}$ (52) and disbalance ratio of input beamsplitters$q$ (71). Panel (a) shows the case of balanced input beamsplitter, while panel (b) corresponds to unbalanced case. All other beamsplitters are assumed to be balanced. Channel parameters: $V = 5$, $T = 0.95$, $\xi = 10^{-3}$.

Next, we compute the Holevo information in the untrusted-noise scenario, in which detection imperfections are modeled as additional noise accessible to the eavesdropper (Eve). We assume that Eve controls a Gaussian channel that modifies only the covariance matrix, leaving the mean unchanged, while Bob's measurements are considered ideal.

$$\mathcal{I}_A \otimes \Phi_N^{(\gamma)} : \Sigma_{\mathrm{AB}}^{\mathrm{EB}} \mapsto \Sigma_{\mathrm{AB}}^{(\gamma)} = \begin{pmatrix} V\mathbb{1} & c\sigma_z \\ c\sigma_z & V_{\mathrm{B}}\mathbb{1} + \Sigma_N^{(\gamma)} \end{pmatrix}, \quad \gamma \in \{\mathrm{H}, \mathrm{DH}\} \tag{85}$$

where $\Sigma_N^{(\mathrm{H})}$ ($\Sigma_N^{(\mathrm{DH})}$) is given by Eq. (33) (Eq. (68)).

Assuming measurement noise as accessible to Eve, her entropy $S_{\mathrm{E}} = S_{\mathrm{AB}}$ is evaluated from symplectic eigenvalues of the covariance matrix (85) given by

$$\nu_{1,2}^{(\gamma)} = \sqrt{\Delta_\gamma/2 \pm \sqrt{\Delta_\gamma^2/4 - D_\gamma}}, \tag{86}$$

$$D_\gamma = \det \Sigma_{\mathrm{AB}}^{(\gamma)}, \ \Delta_\gamma = \det(V_{\mathrm{B}}\mathbb{1} + \Sigma_N^{(\gamma)}) + V^2 - 2c^2 \tag{87}$$

The conditional entropy $S_{\mathrm{E|B}} = S_{\mathrm{A|B}}$ is found via partial measurement formula [30], written for $\Sigma_{\mathrm{AB}}^{(\gamma)}$ as follows:

$$\Sigma_{\mathrm{A|B}}^{(\gamma)} = V\mathbb{1} - c^2\sigma_z(V_{\mathrm{B}}\mathbb{1} + \Sigma_m^{(\gamma)})^{-1}\sigma_z. \tag{88}$$

$$\nu_3^{(\gamma)} = \sqrt{\det \Sigma_{\mathrm{A|B}}^{(\gamma)}} \tag{89}$$

For homodyne detection, calculating symplectic eigenvalue of conditional covariance matrix as square root of the determinant yields:

$$\nu_3^{(\mathrm{H})} = \sqrt{V\left(V - \frac{c^2}{V_{\mathrm{B}} + \sigma_N}\right)} \tag{90}$$

The same calculation for double homodyne results in:

$$\nu_3^{(\mathrm{DH})} = V\sqrt{\frac{\left(V_{\mathrm{B}} + \delta_1 - \frac{c^2}{V}\right)\left(V_{\mathrm{B}} + \delta_2 - \frac{c^2}{V}\right)}{(V_{\mathrm{B}} + \delta_1)(V_{\mathrm{B}} + \delta_2)}}, \tag{91}$$

where $\delta_i = 2\sigma_i - 1$ per Eq. (70). Assuming purification attack, Holevo information is calculated as

$$\begin{aligned} \chi_{\mathrm{EB}}^{(\gamma)} &= S_{\mathrm{E}} - S_{\mathrm{E|B}} = S_{\mathrm{AB}}^{(\gamma)} - S_{\mathrm{A|B}}^{(\gamma)} \\ &= g(\nu_1^{(\gamma)}) + g(\nu_2^{(\gamma)}) - g(\nu_3^{(\gamma)}), \end{aligned} \tag{92}$$

where

$$g(\nu) = \frac{\nu+1}{2}\log\frac{\nu+1}{2} - \frac{\nu-1}{2}\log\frac{\nu-1}{2}. \tag{93}$$

Finally, the asymptotic secret fraction reads

$$K^{(\gamma)} = \beta I_{AB}^{(\gamma)} - \chi_{EB}^{(\gamma)}, \tag{94}$$

where β is the reconciliation efficiency.

Note that in the case of double homodyne detection, joint entropy S_{AB} and thereby Holevo information χ_{EB} are explicitly dependent on squeezing parameter $r \in [r_2, r_1]$ (72), as seen from the expression for covariance matrix (85). Moreover, as illustated by Fig. 6, the choice of r significantly affects the value of χ_{EB}. Regarding Fig. 6, for ideal double homodyne, $\sigma_x^{(1)} = \sigma_x^{(2)} = 1$ (black dot), only singular value of $r = \rho$ (73) is allowed,as shown at point ρ in Fig. 5. If input beansplitter is balanced, $q = 1$ (Fig. 6a), then $r = 0$, and the case of projector onto coherent state (57) is realized. Otherwise, if $q \neq 1$, $r \neq 0$ as well, representing projector onto squeezed state (75). In both cases, the value of S_{AB} is independent of ρ and determined solely by modulation variance V_{A} and channel parameters T and ξ. In the symmetrical but nonideal case (red curve in Fig. 6a), symmetrical interval $[-r_1, r_1]$ arises, and joint entropy is maximized by the midpoint $r = 0$. In this case, choosing the POVM in the form superposition of coherent states (62) is not only allowed, but optimal. If input beamplitter is unbalanced (red curve in Fig. 6b), interval of possible r, as well as optimal squeezing parameterthat maximazis Holevo information, shift. Moreover, if disbalance of input beamsplitter $|1 - q|$ is sufficiently high, choice of $r = 0$ becomes impossible as it does not lay in the interval (72). Substituing $r \notin [r_2, r_1]$ results in one of eigenvalues being less than 1, which is unphysical. Therefore, the more general representation given by Eq. (77) must be employed. In asymmetrical case (blue curves) if input beamsplitter is balanced, while using $r = 0$, i.e. Eq. (62), is possible, joint entropy is maximized by $r \neq 0$. If input beamsplitter is unbalanced, similarly to red curve, dependencies of r shift in accordance to disbalance ratio. For calculations, we will optimize r to maximize χ_{EB}.

The effects of detection asymmetry on mutual information, Holevo information, and the asymptotic secret fraction are illustrated in Fig. 7 for homodyne detection

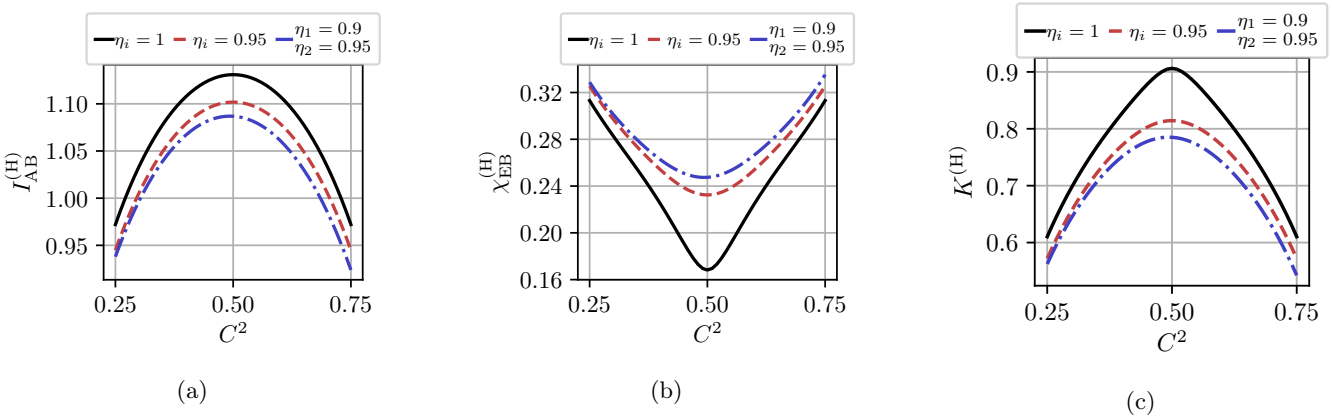


Figure 7: Dependence of (a) mutual information, (b) Holevo information, and (c) asymptotic secret fraction on beam splitter transmission C^2 if receiver uses homodyne detection, for different detector efficiencies at $V = 5$, $T = 0.95$, $\xi = 10^{-3}$, $\beta = 0.95$.

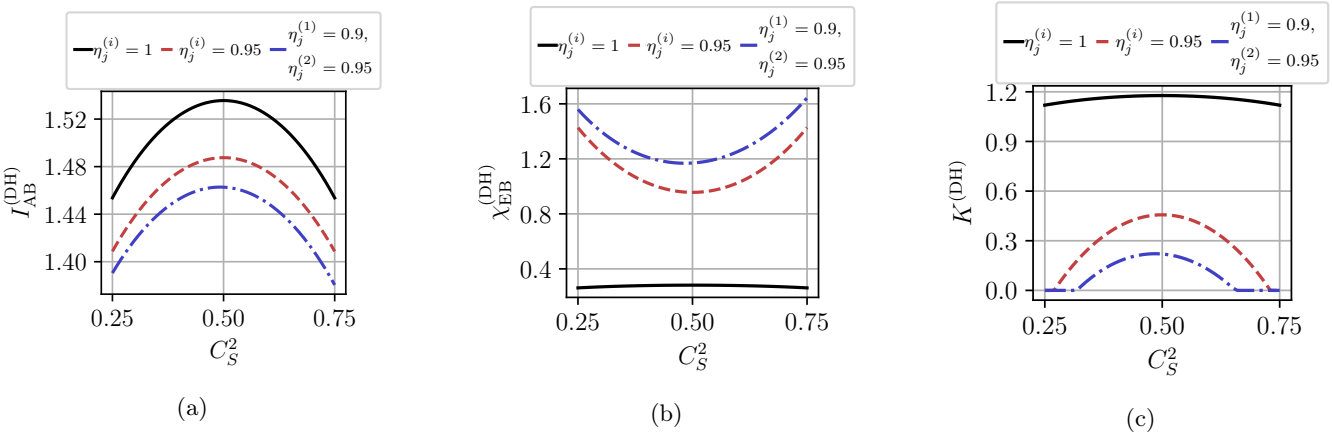


Figure 8: Dependence of (a) mutual information, (b) Holevo information, and (c) asymptotic secret fraction on signal beam splitter transmission C_S^2 (see Fig. 3) if receiver uses double homodyne detection. All other beam splitters are assumed to be balanced. Results are shown for various detector efficiencies; efficiencies not specified in the legend are taken to be unity. Parameters: $V = 5$, $T = 0.95$, $\xi = 10^{-3}$, $\beta = 0.95$.

and Fig. 8 for double homodyne detection. From Fig. 7, it is evident that while asymmetry in homodyne detection generally degrades performance, the combination of an unbalanced beamsplitter and unequal detector efficiencies can be beneficial, resulting in maxima of the mutual information and secret fraction with given parameters. Correspondingly, the minima of the Holevo information exhibit similar behavior. Our numerical results for the homodyne case are qualitatively consistent with Ref. [21], namely the dependence of the asymptotic secret fraction on the beam splitter transmittance deviation and balanced detector deviation. Similar behavior can be seen from Fig. 8 for double homodyne. However, while mutual information in double homodyne is generally higher than in homodyne, the significantly greater increase in Holevo information leads to a lower secret fraction overall. An interesting feature illustrated by the black curves in this figure is that, under ideal detection efficiencies, variations in the beamsplitter transmission cause reduction in the Holevo information. As seen previosly in Fig. 6, this is the case of joint entropy being constant in r and therefore C_S^2 per Eq. (73), so that only dependency of conditional entropy on asymmetry remains. Conditional entropy is reduced by asymmetry, see

The dependence of the asymptotic secret fraction on channel length in the presence of asymmetrical detection is shown in Fig. 9. These results indicate that in the untrusted noise scenario, asymmetry noise significantly reduces maximal channel length. Consistent with previous results, asymmetrical double homodyne performs worse than asymmetrical homodyne, due to Holevo information being high. The discrepancy in our Holevo information calculations leads to results for the asymptotic secret fraction dependence under double homodyne de-

tection that differ from those in Ref. [23], although both works reach the same conclusion regarding its degraded performance.

V. DISCUSSION AND CONCLUSION

In this paper, we have studied the effects of asymmetry introduced by unbalanced beam splitters and different efficiencies of the photodetectors in photocount statistics of homodyne and double homodyne detection. By using the Gaussian approximation, we have developed the method for constructing POVMs of homodyne-based schematics. This method is applied to deduce the expression for the POVM that generalizes the well-known results to the case of the asymmetric homodyne detection. This POVM has been found to be well defined across all parameter settings of the scheme, with the excess noise variance modified by the asymmetry.

Our formalism allows us to easily analyze various homodyne-based schematics. It can be extended to describe more complex measurement systems. We have demonstrated this by performing an analysis for the eight-port asymmetric double homodyne scheme. For this scheme, we have deduced the Gaussian approximation (see Eqs. (45)-(47)) and the corresponding POVM expressed in terms of the incoherent Gaussian superposition of projectors onto coherent states (see Eq. (62)). As shown in Fig. 4, the asymmetry induces effects such as shifts and anisotropy of the distributions in the photocount difference plane. However, the applicability of double homodyne POVM in the form (62) may be broken provided that the beam splitter for the signal mode is unbalanced. To resolve this, an extension to the set of squeezed coherent states is needed (see Eq. (77)), leading to explicit dependency of POVM on the squeezing parameter, which lies within the interval (72), as illustrated in Fig. 5. This implies that, in general, there are infinitely many POVMs representing a single set of parameters for the double homodyne scheme. Consequently, the squeezing parameter requires an additional constraint to ensure the POVM is well-defined. This inherent non-uniqueness becomes relevant in untrusted noise security analysis. Specifically, while mutual information between Alice and Bob in Gaussian modulated coherent states CV-QKD protocol is calculated with variances of quadrature distribution (48) (see Appendix C for explicit calculations), joint entropy, and thus Holevo information, is explicitly dependent on the squeezing parameter (see Fig. 6). We optimized the squeezing parameter to maximize Holevo information.

In this paper we analyzed how measurement asymmetry impacts the performance of CV-QKD system in the untrusted noise scenario, where asymmetry noise is accessible to Eve. In this case even relatively small asymmetry significantly reduces maximum channel length (see Fig. 9). In trusted-noise CV-QKD [24], if Alice and Bob are unaware of detector's arms asymmetry, they may misinterpret the increased variance as channel excess noise rather than trusted detector imperfection. This leads to an overestimation of channel excess noise and an underestimation of trusted detector noise. Similar to blinding attacks [27], this vulnerability can be mitigated by inserting attenuators to balance the detection scheme. Despite the existence of countermeasures for the attacks described, implementing analytical corrections based on the formalism developed in this paper would be preferable for accurate security assessment and performance optimization.

ACKNOWLEDGEMENTS

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Appendix A: Statistical distance between Gaussian approximation and Skellam distribution

In this Appendix, we study the perfomance of Gaussian approximation for the statistics of photon count difference (10):

$$\mathsf{P}_G(\mu) = G(\mu - \mu_G; \sigma_G), \tag{A1}$$

$$\sigma_G = \eta_1|\alpha_1|^2 + \eta_2|\alpha_2|^2 \approx (\eta_1 S^2 + \eta_2 C^2)\,|\alpha_L|^2\,, \tag{A2}$$

$$\mu_G = \eta_1|\alpha_1|^2 - \eta_2|\alpha_2|^2 \approx (\eta_1 S^2 - \eta_2 C^2)\,|\alpha_L|^2 + CS(\eta_1+\eta_2)\,|\alpha_L|\,\langle \hat{x}_\phi\rangle \tag{A3}$$

by comparing it with the exact statistics of difference events governed by the Skellam distribution (7):

$$\mathsf{P}(\mu) = e^{-\eta_1|\alpha_1|^2} e^{-\eta_2|\alpha_2|^2} \left(\frac{\eta_1|\alpha_1|^2}{\eta_2|\alpha_2|^2}\right)^{\mu/2} \times I_\mu\big(2\sqrt{\eta_1\eta_2|\alpha_1|^2|\alpha_2|^2}\big), \tag{A4}$$

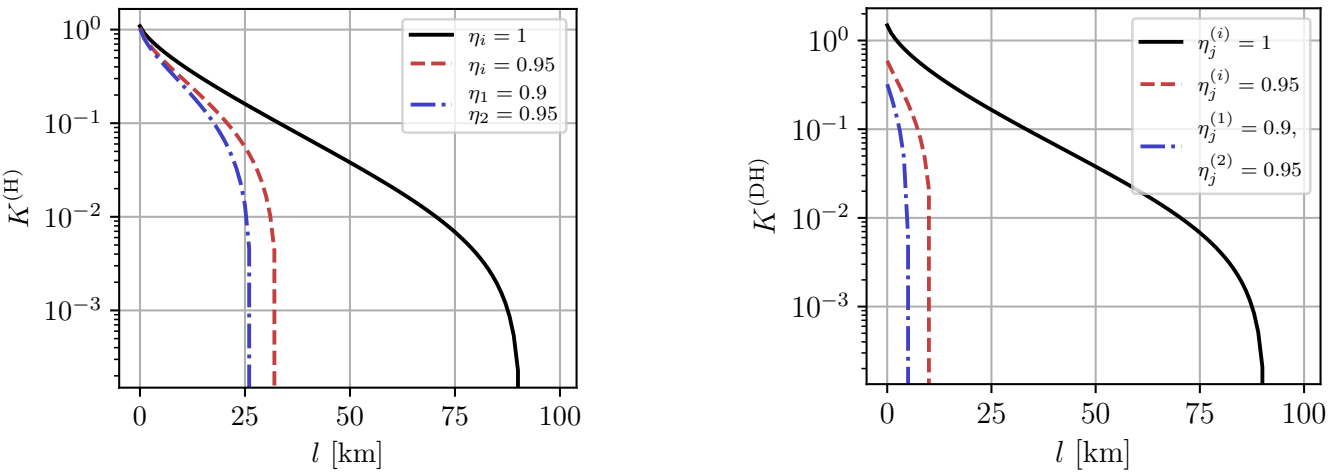
both repeated here for ease. We limit our numerical results to the coherent signal state only, except in cases where the curves show sufficiently distinct differences.

We evaluate the statistical distance between the probability distributions using the total variational distance that can be computed as half of the L^1 distance

$$D_P \equiv \mathtt{D}\,(\mathsf{P}, \mathsf{P}_G) \equiv \frac{1}{2}\sum_{\mu=-\infty}^{\infty} |\mathsf{P}(\mu) - \mathsf{P}_G(\mu)|\,. \tag{A5}$$

Note that, according to Eq. (A5), μ takes integer values and we evaluate the distance between the probability mass fuctions, whereas the normalization condition for the Gaussian function (A1)

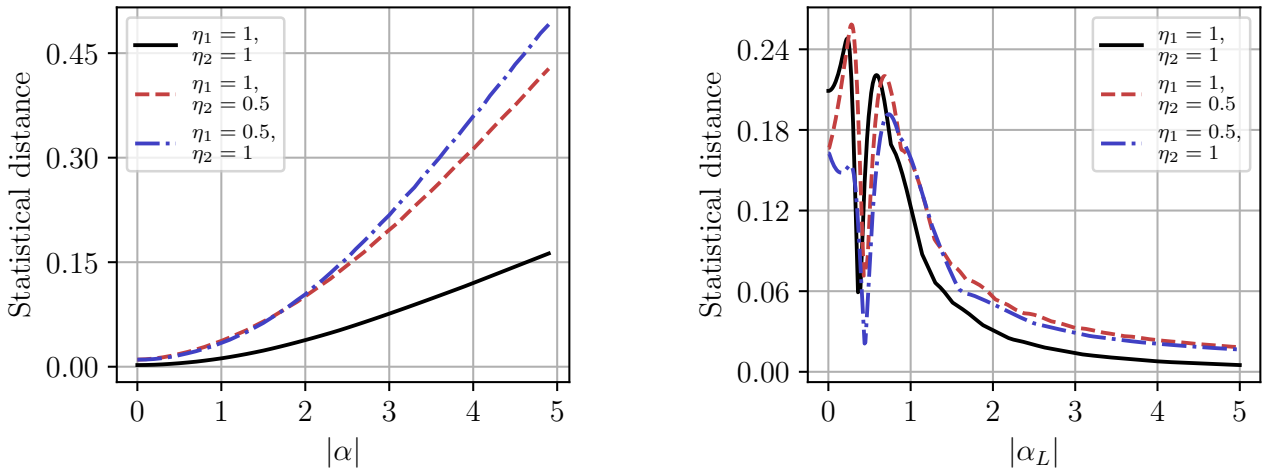
$$\int_{-\infty}^{\infty} \mathrm{d}\mu\, \mathsf{P}_G(\mu) = 1 \tag{A6}$$



(a) Homodyne detection

(b) Double homodyne detection

Figure 9: Asymptotic secret fraction as a function of channel length computed for (a) homodyne detection and (b) double homodyne detection, shown for various detector efficiencies. Parameters are $V = 5, T = 0.95, \xi = 10^{-3}, \beta = 0.95$. The losses are assumed to be 20 dB per 100 km.



(a) $|\alpha_L| = 5$

(b) $|\alpha| = 0.5$

Figure 10: Statistical distance $D_P = \mathtt{D}(\mathsf{P}, \mathsf{P}_G)$ as a function of (a) signal amplitude and (b) LO amplitude at different detector efficiencies.

implies applicability of the continuum limit. For integer μ, the integral on the left hand side of Eq. (A6) should be replaced with a sum and we have the relation

$$\sum_{\mu=-\infty}^{\infty} \mathsf{P}_G(\mu) = \vartheta_3(\pi\mu_G, e^{-2\pi^2\sigma_G}) \equiv N_G \tag{A7}$$

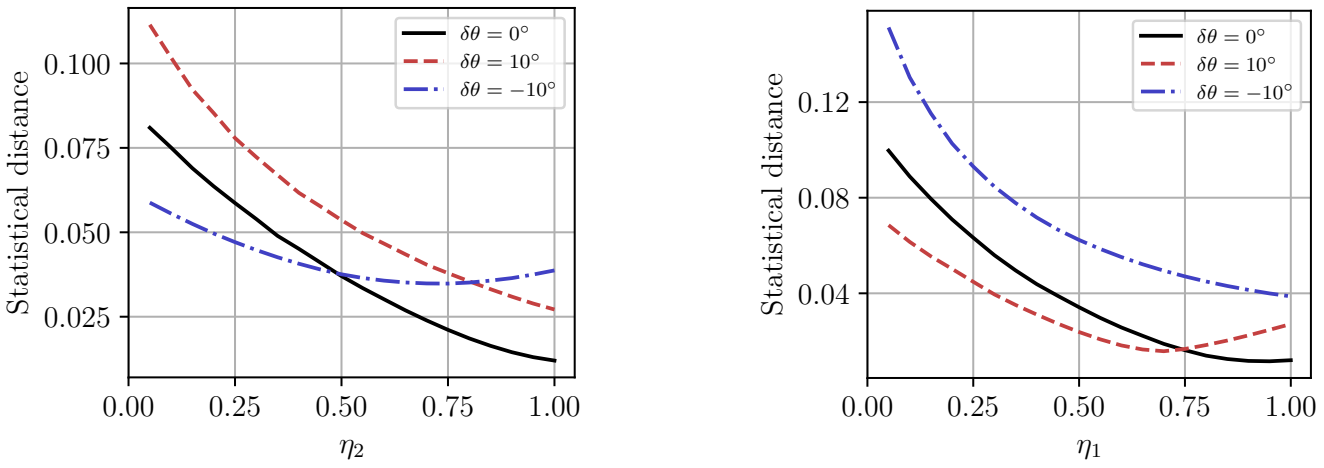
where ϑ_3 is the Jacobi elliptic theta function [33].

In the applicability region of the continuum limit, the normalization constant N_G is close to unity. The numerical analysis shows that $|N_G - 1| \leq 10^{-4}$ at $2\sigma_G \geq 1$. The latter gives the condition for the LO amplitude

$$|\alpha_L| \geq \frac{1}{\sqrt{2(\eta_1 S^2 + \eta_2 C^2)}} \equiv \alpha_N \tag{A8}$$

which ensures both applicability of the continuum limit and proper normalization of the Gaussian approximation. In our calculations, the probability P_G will be numerically corrected by introducing the factor N_G^{-1} provided that $|\alpha_L|$ is below the "renormalization point" α_N.

The curves presented in Fig. 10 illustrate how the accuracy of the Gaussian approximation is affected by the signal and LO amplitudes, $|\alpha|$ and $|\alpha_L|$. More specifi-



(a) $\eta_1 = 1$ (b) $\eta_2 = 1$

Figure 11: Statistical distance $D_P = \mathtt{D}(\mathsf{P}, \mathsf{P}_G)$ as a function of photodetector efficiency (a) η_2 at $\eta_1 = 1$ and (b) η_1 at $\eta_2 = 1$ for different values of the beam splitter disbalance angle $\delta\theta$ (see eq. (A9)). The amplitudes are $|\alpha| = 1$ and $|\alpha_L| = 5$.

cally, in Fig. 10a (Fig. 10b), the statistical distance is numerically evaluated as a function of the amplitude $|\alpha|$ ($|\alpha_L|$) at different values of the photodetectors efficiencies provided that the value of the other amplitude $|\alpha_L|$ ($|\alpha|$) is fixed.

Referring to Fig. 10a, the curves behave as expected: given the LO amplitude $|\alpha_L|$, the distance monotonically increases with $|\alpha|$. It is shown that, at $|\alpha_L| = 5$ and $|\alpha| > 1$, the perfectly symmetric homodyne presents the case with minimal distance, D_P, while in the presence of asymmetry, the curves exhibit a rapid growth and the quality of the Gaussian approximation rapidly degrades to the point, where $D_P > 0.1$, so it is not useful for its intended purpose.

When it comes to dependencies of the statistical distance on the LO oscillator amplitude computed at fixed value of $|\alpha|$, the above results suggest that the smaller the amplitude $|\alpha|$ the better the accuracy of the Gaussian approximation. We can also expect the distance will be small provided that $|\alpha_L|$ is large and the strong-LO approximation is applicable.

Referring to Fig. 10b, the curves evaluated at $|\alpha| = 0.5$ display a non-monotonic behavior with two local maxima in the weak LO range where $|\alpha_L| < 1$. By contrast, after the second maximum at $|\alpha_L| > 1$, the distance falls with the LO amplitude and it drops below 0.05 at $|\alpha_L| > 2$.

Note, that, when $|\alpha| < 0.1$ and the signal mode state is close to the vacuum state, the two local maxima of D_P can be estimated to be slightly above 0.08 and 0.1, respectively. So, in this case, the distribution (10) might be regarded as a reasonable approximation even in the weak LO range where the probability P_G approaches the close neighborhood of the singular limit, $\lim_{|\alpha_L|\to 0} \mathsf{P}_G(\mu) = \delta(\mu)$.

From Fig. 10b, it can also be seen that the distance vs LO amplitude dependence that can be used as a tool to characterize the applicability region of the strong-LO approximation is nearly insensitive to asymmetry. In other words, the latter does not produce noticeable effects on the accuracy of the approximation.

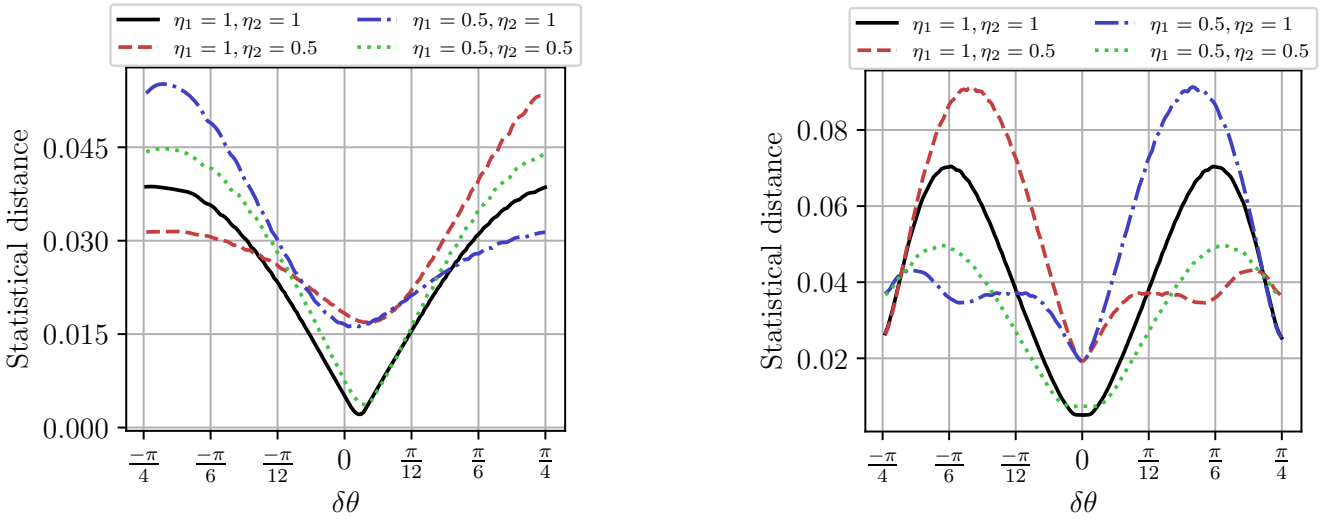
The parameters describing the photodetection asymmetry are the efficiencies η_1 and η_2. The curves plotted in Fig. 10 are computed at different values of the efficiencies.

To quantify deviation of the beam splitter transmission and reflection amplitudes from the balanced $50:50$ values $t = \cos\theta = r = \sin\theta = 1/\sqrt{2}$ at the angle $\theta = \pi/4$, we introduce the beam splitter *disbalance angle* given by

$$\delta\theta \equiv \frac{\pi}{4} - \theta. \tag{A9}$$

In Figure 11 we plot the distance against the efficiency of the photodetector assuming that the other photodetector is perfect. The curves are evaluated at different values of the disbalance angle (A9).

In Fig. 12 the statistical distance vs disbalance angle curves are presented for coherent and one-photon signal states. These curves illustrate how the beam splitter disbalance and the photodetector efficiencies influence the accuracy of the Gaussian approximation. The distance is shown to be minimal in the vicinity of the balanced beam splitter point with a vanishing disbalance angle, $\delta\theta = 0$. The efficiency dependence of the distance is shown to decrease monotonically at $\delta\theta = 0$. In contrast, for disbalanced beam splitter, this dependence can reveal non-monotonic behavior.



(a) $|\psi\rangle = |\alpha\rangle$, $\alpha = 0.5$

(b) $|\psi\rangle = |n\rangle$, $n = 1$

Figure 12: Statistical distance $D_P = \mathtt{D}(\mathsf{P}, \mathsf{P}_G)$ as a function of the beam splitter disbalance angle, $\delta\theta$, for the signal mode prepared in (a) the coherent state and in (b) the single photon Fock state at different efficiencies with $|\alpha_L| = 5$.

Appendix B: Gaussian approximation from Skellam distribution

The derivation procedure for the Gaussian approximation outlined in Sec. II transforms the photocount difference probability (6) into the form of a convolution of the normal distributions by approximating the Poisson distributions. In this Appendix, we discuss an alternative method where the starting point is the Skellam distribution (7). For convenience, we shall reproduce the expression for this distribution here:

$$P(\mu) = e^{-\eta_1|\alpha_1|^2} e^{-\eta_2|\alpha_2|^2} \left(\frac{\eta_1|\alpha_1|^2}{\eta_2|\alpha_2|^2}\right)^{\frac{\mu}{2}} \times I_\mu\left(2\sqrt{\eta_1\eta_2|\alpha_1|^2|\alpha_2|^2}\right), \tag{B1}$$

where $I_k(z)$ is the modified Bessel function of the first kind and the amplitudes $|\alpha_{1,2}|$ are given by Eq. (3).

The method under consideration (see, e.g. the textbook [9]) assumes that, in the strong LO limit, the argument of the modified Bessel function is large and $I_\mu(z)$ can be approximated using its asymptotic expansion taken in the Gaussian form:

$$I_\mu(z) \approx \frac{1}{\sqrt{2\pi z}} \exp\left[z - \frac{\mu^2}{2z}\right]. \tag{B2}$$

This formula can be deduced by performing a saddle-point analysis for the integral representation of the Bessel functions [37].

Heuristically, it can also be obtained from from the lowest order asymptotic expansion for the Bessel function [33]: $I_\mu(z) \approx e^z(1-(4\mu^2-1)/(8z))/\sqrt{2\pi z}$ assuming that, for small values of x, $1-x$ can be replaced with e^{-x} (the factors independent of μ are not essential because they can be incorporated into the normalization factor of the Gaussian approximation).

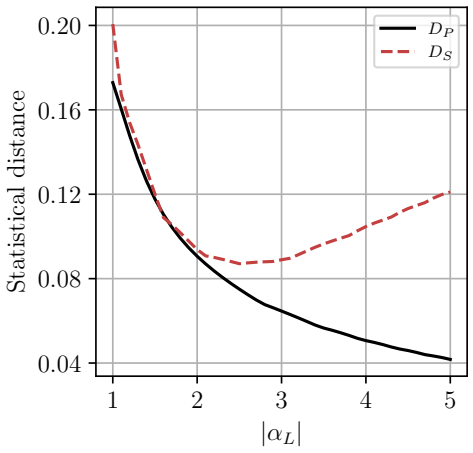


Figure 13: Distances, $D_P = D(\mathsf{P}, \mathsf{P}_G)$ and $D_S = D(P, \mathsf{P}_G^{(s)})$, between the Skellam distribution, P, and two Gaussian approximations, P_G (see Eq. (10)) and $\mathsf{P}_G^{(s)}$ (see Eq. (B5)), as a function of $|\alpha_L|$ the beam splitter disbalance angle at $\delta\theta = 15^\circ$ $\alpha = 1$, and $\eta_1 = \eta_2 = 1$.

Assuming that $CS \neq 0$ and $|\alpha_L|$ is sufficiently large, we can use the approximate relations

$$z = 2\sqrt{\eta_1\eta_2}|\alpha_1||\alpha_2| \approx 2CS\sqrt{\eta_1\eta_2}\,|\alpha_L|^2, \tag{B3}$$

$$\ln\left(\frac{\eta_1|\alpha_1|^2}{\eta_2|\alpha_2|^2}\right)^{\frac{\mu}{2}} \approx \frac{\mu}{2}\left(\ln\frac{\eta_1 S^2}{\eta_2 C^2} + \frac{\langle\hat{x}_\phi\rangle}{CS\,|\alpha_L|}\right) \tag{B4}$$

to obtain the Gaussian approximation for the Skellam distribution (B1) given by

$$\mathsf{P}_G^{(s)}(\mu) = G(\mu - \tilde{\mu}_G; \tilde{\sigma}_G), \tag{B5}$$

$$\tilde{\mu}_G = \sqrt{\eta_1\eta_2}\Big[CS\,|\alpha_L|^2 \ln\frac{\eta_1 S^2}{\eta_2 C^2} + |\alpha_L|\,\langle\hat{x}_\phi\rangle\Big], \tag{B6}$$

$$\tilde{\sigma}_G = 2CS\sqrt{\eta_1\eta_2}\,|\alpha_L|^2\,. \tag{B7}$$

Similar to Eq. (15), we cast the probabilty (B5) into the following quadrature form:

$$\mathsf{P}_G^{(s)}(\tilde{x}) = \frac{1}{\sqrt{2\pi\tilde{\sigma}_G}}\exp\Big\{-\frac{(\tilde{x} - \langle\hat{x}_\phi\rangle)^2}{2\tilde{\sigma}_x}\Big\}, \tag{B8}$$

$$\tilde{x} = \frac{\mu}{\sqrt{\eta_1\eta_2}\,|\alpha_L|} - CS\,|\alpha_L|\ln\frac{\eta_1 S^2}{\eta_2 C^2}, \tag{B9}$$

$$\tilde{\sigma}_x = \frac{2CS}{\sqrt{\eta_1\eta_2}}, \tag{B10}$$

so that we may follow the line of reasoning presented in Sec. II to deduce the POVM

$$\hat{\Pi}_G^{(s)} = \frac{1}{\sqrt{\eta_1\eta_2}\,|\alpha_L|} \times \int \mathrm{d}x' G(x - x'; \tilde{\sigma}_N)|x', \phi\rangle\langle x', \phi| \tag{B11}$$

with the noise variance

$$\tilde{\sigma}_N = \tilde{\sigma}_x - 1, \quad 0 \le \tilde{\sigma}_x \le \tilde{\sigma}_x^{(\max)} = 1/\sqrt{\eta_1\eta_2}. \tag{B12}$$

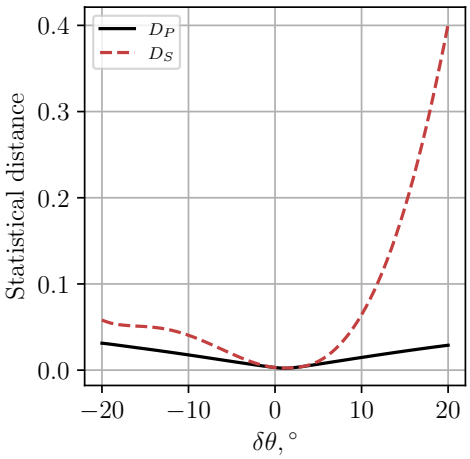


Figure 14: Distances, $D_P = D(\mathsf{P}, \mathsf{P}_G)$ and $D_S = D(\mathsf{P}, \mathsf{P}_G^{(s)})$, between the Skellam distribution, P, and two Gaussian approximations, P_G (see Eq. (10)) and $\mathsf{P}_G^{(s)}$ (see Eq. (B5)), as a function of the beam splitter disbalance angle $\delta\theta$ at $\alpha = 1$, $\alpha_L = 10$ and $\eta_1 = \eta_2 = 1$.

Note that, in the symmetric case with $C = S$ and $\eta_1 = \eta_2$, the Gaussian distributions given by Eqs. (10) and (B5) are equivalent. This is no longer the case in the presence of asymmetry effects.

Figure 13 demonstrates that, at $\delta\theta \neq 0$, by contrast to the distance between P and P_G, $D_P = \mathtt{D}(\mathsf{P}, \mathsf{P}_G)$ which is a monotonically decreasing function of $|\alpha_L|$, the distance between the Skellam distribution and the approximate distribution (B5), $D_S = \mathtt{D}(\mathsf{P}, \mathsf{P}_G^{(s)})$, reveals non-monotonic behaviour and increases with $|\alpha_L|$ at sufficiently large LO amplitudes. Referring to Fig. 14, disbalance of the beam splitter has strong detrimental effect on the accuracy of the approximation (B5).

What is more important is that, by contrast the noise excess variance (34) which is always positive, the variance (B12) becomes negative when $2CS \le \sqrt{\eta_1\eta_2}$. The latter breaks applicability of Eq. (B11) giving an ill-posed POVM.

Appendix C: Mutual information and input state of entanglement-based scheme

In this appendix, we begin with derivation of the expressions for the mutual information between Alice and Bob in the Gaussian-modulated coherent-state CV-QKD protocol discussed in Sec. IV. These expressions are given by Eqs. (80) and (81).

The important point greatly simplifying calculations is that both the channel (79) and the measurements are Gaussian. As a result, the random variables that determine the mutual information appear to be normally distributed, so that we can use the well-known analytical result that expresses the mutual information of the Gaussian random variables in terms of the determinats of the suitably defined covariance matrices (see, e.g., calculations detailed in online book [38]).

Specifically, transformations that a coherent state undergoes under the action of the channel $\mathcal{E}_{A\to B}$ can be explicitly represented as follows

$$|\alpha\rangle\langle\alpha| \xrightarrow{\mathcal{E}_T} |\sqrt{T}\alpha\rangle\langle\sqrt{T}\alpha| \xrightarrow{\Phi_\xi} N_\xi \int \mathrm{d}^2\beta \exp\Big\{-\frac{2|\beta|^2}{\xi}\Big\} \times |\sqrt{T}\alpha + \beta\rangle\langle\sqrt{T}\alpha + \beta|, \tag{C1}$$

where $N_\xi = 2/(\pi\xi)$. When Bob performs a Gaussian measurement described by the POVM $\{\hat{\Pi}_G^{(\gamma)}(x)\}_x$, where the value of x labels the measurement outcomes (e.g., for homodyne detection, x is the quadrature variable given by Eq. (16)), the joint probability density can be computed as follows

$$p_{\mathrm{B}}^{(\gamma)}(x, \alpha) = p_B^{(\gamma)}(x|\alpha)p_A(\alpha), \quad \gamma \in \{\mathrm{H}, \mathrm{DH}\} \tag{C2a}$$

$$\begin{aligned} p_B^{(\gamma)}(x|\alpha) &= \mathrm{Tr}\{\mathcal{E}_{A\to B}(|\alpha\rangle\langle\alpha|)\hat{\Pi}_G^{(\gamma)}(x)\} \\ &= \mathrm{Tr}\{\mathcal{E}_T(|\alpha\rangle\langle\alpha|)\Phi_\xi^*(\hat{\Pi}_G^{(\gamma)}(x))\} \\ &= \langle\sqrt{T}\alpha|\Phi_\xi(\hat{\Pi}_G^{(\gamma)}(x))|\sqrt{T}\alpha\rangle, \end{aligned} \tag{C2b}$$

where we have used the fact that the additive noise channel $\hat{\Phi}_\xi$ is self-dual: $\hat{\Phi}_\xi^* = \hat{\Phi}_\xi$.

For homodyne detection, we can combine Eq. (C2) and the expression for the Q-symbol of POVM given by Eq. (15) to deduce the joint probability density

$$p_{\rm B}^{(\rm H)}(x,\tilde{\alpha}_1) \propto \exp\left[-\frac{(x-2\tilde{\alpha}_1)^2}{2(\sigma_x+\xi)} - \frac{2\tilde{\alpha}_1^2}{TV_{\rm A}}\right] = \exp\Big\{-\frac{1}{2}\begin{pmatrix}\tilde{\alpha}_1 & x\end{pmatrix}(\Sigma_{\rm H})^{-1}\begin{pmatrix}\tilde{\alpha}_1\\ x\end{pmatrix}\Big\}, \tag{C3}$$

where

$$\tilde{\alpha}_1 = \sqrt{T}\,\mathrm{Re}\,\alpha e^{-i\phi}, \quad \tilde{\alpha}_2 = \sqrt{T}\,\mathrm{Im}\,\alpha e^{-i\phi}. \tag{C4}$$

The covariance matrix of the normal distribution (C3)

$$\Sigma_{\rm H} = \begin{pmatrix}\frac{TV_{\rm A}}{4} & \frac{TV_{\rm A}}{2}\\ \frac{TV_{\rm A}}{2} & TV_{\rm A}+\sigma_x+\xi\end{pmatrix} \tag{C5}$$

gives the mutual information between Alice and Bob that can be computed from the formula [38]

$$I_{\rm AB}^{(\rm H)} = \frac{1}{2}\log\frac{\Sigma_{11}^{\rm H}\Sigma_{22}^{\rm H}}{\det\Sigma_{\rm H}} = \frac{1}{2}\log\left[1+\frac{TV_{\rm A}}{\sigma_x+\xi}\right], \tag{C6}$$

where $\Sigma_{ii}^{\rm H}$ are the diagonal elements of $\Sigma_{\rm H}$ and all logarithms are base 2.

Similarly, for double homodyne detection, from Eqs. (C2) and (48) we have

$$p_{\rm B}^{(\rm DH)}(x,\tilde{\alpha}) \propto \prod_{i=1}^{2}\exp\left[-\frac{(x_i-\tilde{\alpha}_i)^2}{\sigma_i+\xi/2} - \frac{2\tilde{\alpha}_i^2}{TV_{\rm A}}\right] = \prod_{i=1}^{2}\exp\Big\{-\frac{1}{2}\begin{pmatrix}\tilde{\alpha}_i & x_i\end{pmatrix}(\Sigma_{\rm DH}^{(i)})^{-1}\begin{pmatrix}\tilde{\alpha}_i\\ x_i\end{pmatrix}\Big\}. \tag{C7}$$

Analogously to Eq. (C5), we obtain the diagonal blocks of the covariance matrix $\Sigma_{\rm DH}$ given by

$$\Sigma_{\rm DH}^{(i)} = \begin{pmatrix}\frac{TV_{\rm A}}{4} & \frac{TV_{\rm A}}{4}\\ \frac{TV_{\rm A}}{4} & (TV_{\rm A}+2\sigma_i+\xi)/4\end{pmatrix}, \quad i\in\{1,2\}. \tag{C8}$$

Since $p_{\rm B}^{(\rm DH)}(x,\tilde{\alpha}) = p_{\rm B}^{(\rm DH)}(x_1,\tilde{\alpha}_1)p_{\rm B}^{(\rm DH)}(x_2,\tilde{\alpha}_2)$, the mutual information can be evaluated as a sum of two terms, determined by the above formula (C6) with $\Sigma_{\rm H}$ replaced by $\Sigma_{\rm DH}^{(i)}$. The result reads

$$I_{\rm AB}^{(\rm DH)} = \frac{1}{2}\sum_{i=1,2}\log\frac{[\Sigma_{\rm DH}^{(i)}]_{11}[\Sigma_{\rm DH}^{(i)}]_{22}}{\det\Sigma_{\rm DH}^{(i)}} = \frac{1}{2}\sum_i\log\left[1+\frac{TV_{\rm A}}{2\sigma_i+\xi}\right]. \tag{C9}$$

In conclusion of this appendix, we briefly discuss equivalence relation between the prepare-and-measure (PM) and the entanglement-based (EB) schemes. To this end, we consider the two mode squeezed vacuum (TMSV) state given by [39]

$$|\Psi_{\rm AB}\rangle = \frac{1}{\cosh r}\sum_{n=0}^{\infty}(\tanh r)^n|n,n\rangle_{\rm AB}, \tag{C10}$$

where r is the real-valued squeezing parameter, with the covariance matrix [30]

$$\Sigma_{\rm TMSV} = \begin{pmatrix}\cosh(2r)\mathbb{1} & \sinh(2r)\sigma_z\\ \sinh(2r)\sigma_z & \cosh(2r)\mathbb{1}\end{pmatrix}. \tag{C11}$$

In the EB protocol, Alice is assumed to hold the state $\hat{\rho}_{\rm AB} = |\Psi_{\rm AB}\rangle\langle\Psi_{\rm AB}|$. Then Alice measures one mode using a noiseless double homodyne measurement, which corresponds to the coherent-state POVM (see Eq. (57)) $\{\hat{\Pi}_{CS}(\beta) = \pi^{-1}|\beta\rangle\langle\beta|\}_\beta$, and sends the second mode to Bob. Unconditional state on the Bob's side prepared by such a measurement is

$$\hat{\rho}_{\rm B}^{\rm EB} = \int \mathrm{d}^2\beta\,\mathrm{Tr}_A\big\{(\hat{\Pi}_{CS}(\beta)\otimes\hat{I}_B)\hat{\rho}_{AB}\big\}. \tag{C12}$$

By using the relation

$$\cosh r\langle\beta|\Psi_{\rm AB}\rangle = \exp\Big\{-\frac{|\beta|^2}{2\cosh^2 r}\Big\}|\tilde{\beta}\rangle, \tag{C13}$$

where $\tilde{\beta} = (\tanh r)\beta^*$, the above density matrix (C12) can be recast into the form

$$\hat{\rho}_{\rm B}^{\rm EB} = \frac{1}{\pi\sinh^2 r}\int \mathrm{d}^2\tilde{\beta}\exp\Big\{-\frac{|\tilde{\beta}|^2}{\sinh^2 r}\Big\}|\tilde{\beta}\rangle\langle\tilde{\beta}|. \tag{C14}$$

When

$$\sinh^2 r = \frac{V_A}{2} \tag{C15}$$

and $\tilde{\beta}$ is changed to α, the expressions for the input density matrix in the PM picture (78) and for the density matrix after Alice's measurement in the EB picture (C14) become identical. Since relation $\cosh(2r) = 1+V_A \equiv V$ immediately follows from Eq. (C15), the covariance matrix (C11) equals the matrix $\Sigma_{\rm TMSV}$ that enters Eq. (82). This results justify using the covariance matrix of TMSV for calculation of the Holevo information in the PM protocol.

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